

# Product Requirements Document — NativelD ROS

**Product:** NativelD ROS (Restaurant Operating System) **Internal codename:** Haven 11 / 702 ROS **Document version:** 1.23 **Last updated:** 2026-05-24 **Status:** Living document — describes the current prototype and the target product **Last updated:** 2026-05-23 **Owner:** Product **Source of truth:** haven11-nextjs/ (Next.js application)

## 1. Document control

| Field             | Value                                                                   |
|-------------------|-------------------------------------------------------------------------|
| Audience          | Product, Engineering, Design, QA, founding operators                    |
| Related artefacts | CLAUDE.md (engineering guide), application source under haven11-nextjs/ |
| Review cadence    | Update on every module-level change                                     |

### 1.1 Revision history

| Version | Date       | Summary                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| 1.0     | 2026-05-22 | Initial PRD, reverse-engineered from the working prototype.                                                                                                                                                                                                                                                                                                                                                               |
| 1.1     | 2026-05-22 | Implemented the 2026-05-22 stakeholder review: module renames ( <b>Front of House</b> , <b>Front of House on Shift</b> , <b>Procurement</b> ), order <b>On hold / Closed</b> statuses, the payment-completes-order rule, shift history & configurable shift periods, and a <b>Juice Bar</b> inventory section. All changes are live in the application. Detail in §20.                                                    |
| 1.3     | 2026-05-22 | Multi-location inventory (branch Main Store + Kitchen / Bar / Juice Bar sub-stores, one shared catalogue), internal stock requests, Strong Room issue receipts, purchase-order management approval, and a 7th <b>Accountant</b> role approving petty cash. Detail in §21.                                                                                                                                                 |
| 1.4     | 2026-05-23 | Staff lifecycle — <b>onboard / offboard</b> (status retained, never deleted), <b>weekly scheduling</b> (Mon–Sun shift assignment), and a new <b>HR Dashboard</b> with KPIs, today's roster, and the week grid. Detail in §22.                                                                                                                                                                                             |
| 1.5     | 2026-05-23 | UX pass — new <b>HR Officer</b> role (own login), <b>file upload</b> for compliance documents (guarantor / contract / ID card), redesigned <b>schedule editor</b> (chip picker + quick-fill presets, à la Deputy / When I Work), and a lower-density list/dashboard layout for /staff and /hr. Detail in §23.                                                                                                             |
| 1.6     | 2026-05-23 | Kitchen prep workflow — <b>dual-unit tracking</b> on inventory items (e.g. yam in kg + pcs), <b>photo capture</b> for waste records (peels, breakages), <b>"Prep trim / peel"</b> waste reason, and a <b>Record waste</b> button on Kitchen home. Detail in §24.                                                                                                                                                          |
| 1.7     | 2026-05-23 | UI-level <b>no-duplicates safeguard</b> on the inventory catalogue: the <b>New product</b> dialog is disabled on Kitchen / Bar / Juice Bar tabs (sub-stores are stocked from the Main Store, never define new SKUs) and blocks duplicate names case-insensitively across the catalogue, pointing the user to the existing record. The dialog also exposes the optional <b>alt unit</b> at creation time. Detail in §24.1. |

| Version | Date       | Summary                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| 1.8     | 2026-05-23 | <b>No cross-branch requesting</b> — the PO "Deliver to" dropdown is replaced with a read-only label showing the current branch. A PO always delivers to the branch the user is viewing; to order for another branch they switch branches first. Combined with transfers (already pinned to current branch) and internal stock requests (already branch-scoped), no path remains for one branch to request products into another. See FR-PO-9.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 1.9     | 2026-05-23 | <b>Categories + bulk-request picker.</b> New <code>category</code> field on every product (14 values from Protein to Cleaning), filterable in the inventory list, shown as a clickable badge per row, and matched by the free-text search. The Internal Stock Request modal is rewritten as a <b>multi-select checklist</b> — search box, category chips, "tick all visible" bulk action, per-row checkbox + qty — so a 50-item request is one screen of work instead of fifty "+ Add row" clicks. See FR-INV-12 / FR-INV-13. Schema bump ( <code>v17</code> ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 1.10    | 2026-05-23 | <b>Categories are now manageable at runtime.</b> Promoted from a hardcoded <code>const</code> to <code>state.inventoryCategories</code> with <code>addCategory</code> / <code>renameCategory</code> / <code>removeCategory</code> actions (rename cascades to every product; delete blocked while in use). A focused <b>Manage categories</b> modal is reached from two unobtrusive entry points — the inventory <b>Filter popover footer</b> and a <code>Manage...</code> link beside the Category dropdown in the <b>New product</b> dialog — both owner/manager-gated. No new nav item. See FR-INV-14. Schema bump ( <code>v18</code> ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 1.11    | 2026-05-23 | <b>Transfer actions are now context-aware.</b> Approve / Reject / Issue only render when the viewer is currently in the <b>source</b> branch (the Strong Room for branch requests); the destination side sees an italic status hint instead of action buttons. This prevents a branch from rejecting (or approving, or issuing) its own request. Receive remains the destination-side action; Receipt is visible to both. See FR-TRF-6.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 1.12    | 2026-05-23 | <b>Strong Room transfer-request modal upgraded to the multi-select picker</b> — same UX as the internal stock-request picker (search, category chips, tick-and-qty, grouped list, " <i>Tick all visible</i> ", live ready/total counts). Each row shows live Strong Room availability so the requester knows what's actually in stock. Same seamless flow for a 50-item request as for an internal sub-store request. See FR-TRF-2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 1.13    | 2026-05-23 | <b>Customer accounts — wallet + house tab + statements.</b> Two-ledger model on every <code>Customer</code> (prepaid <code>wallet</code> , A/R <code>credit</code> capped by <code>creditLimit</code> ), append-only <code>CustomerLedgerEntry</code> , statement generation ( <code>CustomerInvoice</code> with auto-numbered IDs and reminder log), and the full A/R workflow on <code>/customers</code> — KPIs, aging buckets, tabbed customer 360 with Account section, top-up / charge / payment / statement modals, browser-printable invoices with a 5-step reminder cadence. Hotel/PMS-style separation of liability vs. receivable. POS payment chips queued for Phase 2 (data model already ready). See §25. Schema bump ( <code>v19</code> ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 1.14    | 2026-05-23 | <b>UX simplification + RBAC tightening pass.</b> " <i>One click, one task</i> " principle applied throughout. (a) <b>Customer 360 rewritten</b> — no more tabs, no dual-mode toggles. Single-screen layout, <i>Last order</i> card front-and-centre, ONE plain-language balance card (" <i>Has money on file</i> " / " <i>Owes you</i> ") with one primary CTA, big <i>Order history</i> / <i>Account history</i> buttons, advanced actions (manual charge, credit limit, tier) hidden behind a gear-icon menu. (b) <b>Inventory AdjustModal split</b> into two single-task modals ( <code>ReceiveStockModal</code> , <code>StockCountModal</code> ); row link is now <i>Receive</i> · <i>Count</i> — two distinct verbs, no in-modal mode switch. (c) <b>Reusable DateRangePicker</b> (preset chips + custom range) applied to <i>Order history</i> and <i>Account history</i> . (d) <b>Expense disburse / top-up role-gated</b> — cashier can request petty cash but only Accountant / Owner can disburse cash or top up the wallet float; reconciliation restricted to the original requester or the cash custodian. (e) <b>Transfer approve/reject now role-gated AND branch-gated</b> — owner/manager AND at source. Storekeeper at the Strong Room can still <i>Issue</i> but sees "Awaiting management approval" instead of approve/reject. Closes the " <i>cashier can disburse</i> " and " <i>manager dashboard still sees reject</i> " loops. |

| Version | Date       | Summary                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| 1.15    | 2026-05-23 | <b>Suggested restock — manager-in-the-loop auto-replenishment.</b> Toast / Lightspeed pattern. New <code>suggestedRestockQty(item)</code> helper (Toast/NRA rule of thumb: target = $\text{reorder} \times 2$ , capped at source availability, rounded to a sensible step). When a branch Main Store or sub-store has items below par, the inventory page shows an amber banner — " <i>3 items running low — restock from Strong Room?</i> " — that opens the existing multi-select request picker with the low items pre-ticked and quantities pre-filled. The user reviews, adjusts, submits; the existing approval / issue / receive flow runs unchanged. The <code>NewTransferModal</code> was extracted to <code>components/</code> so both <code>/transfers</code> (manual) and <code>/inventory</code> (suggested) share it; both modals now accept a <code>prefill</code> prop. <b>System suggests, human submits</b> — no silent auto-orders. See FR-RPL.                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 1.16    | 2026-05-23 | <b>Performance measurement — staff scorecards + manager scorecards.</b> New <code>lib/performance.ts</code> pure-functions module computes role-weighted scorecards ( <code>scoreStaff</code> , <code>scoreBranch</code> ). <code>/staff</code> page gains a <i>Staff performance</i> section with date-range filter, role filter, top-performer / needs-support headlines, and per-staff cards showing role-relevant metrics (sales · voids · over-pours · waste · attendance · feedback). <code>/reports</code> page gains an <b>owner-only Manager scorecards</b> section ranking branches on revenue · integrity · waste · customer mood · complaints. Industry KPI weighting (Toast / NRA / Lightspeed). Date range powered by the reusable <code>DateRangePicker</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 1.17    | 2026-05-23 | <b>Rider 360, bike P&amp;L, customer-aware delivery flow, settings/floor-plan management, full-product audit (§26).</b> (a) <code>Rider</code> extended with phone, next-of-kin, licence, bike plate / make / acquisition cost; new <code>FleetTxn</code> append-only ledger; on <code>completeDelivery</code> the fee auto-credits the rider's bike P&L; new <code>RiderModal</code> shows identity + Income / Outflow / Net P&L cards + delivery KPIs (on-time rate, avg duration) + ledger feed with a date-range filter. (b) <code>Customer</code> extended with <code>address</code> + <code>addressLandmark</code> ; <b>POS Takeout/Delivery modal rewritten</b> with phone-first lookup, live CRM suggestions, "Returning customer" pill with tier + lifetime stats + last address, and writes a new address back to the CRM on submit — a fresh cashier never has to ask a regular for their address. (c) New <code>/settings</code> page (owner/manager) with <b>table floor-plan management</b> (add / rename / change seats / delete, blocked while occupied, grouped by zone). (d) <b>PRD §26 — Whole-product audit and onboarding roadmap</b> documents every gap surfaced and the phased close. Schema bump ( v20 ).                                                                                                                                                                                                   |
| 1.18    | 2026-05-24 | <b>Payroll itemisation + Staff welfare module.</b> (a) <b>Payslip schema</b> rewritten — every non-statutory line is now an itemised <code>PayslipAdjustment</code> carrying <i>category</i> + <i>reason</i> + <i>amount</i> (e.g. " <i>Performance bonus · Hit November sales target (110% of plan) · +£25,000</i> "). New <code>PayrollAdjustment</code> queue lets a manager add bonuses or deductions <b>between runs</b> with reasons that print on the payslip. <code>runPayroll</code> auto-generates lines from lateness, cash shortages, and welfare-advance instalments — each with the source noted, so an auditor / employee always knows <i>why</i> their pay changed. Adjustments are auto-marked consumed when a run uses them. (b) <b>New <code>/welfare</code> page + module</b> for staff support (medical, bereavement, family emergency, education, wedding, <b>loan / salary advance</b> , gift, other). Lifecycle: Pending → Approved → Disbursed → Closed / Repaying → Closed; rejection captures a reason. <b>Salary advances feed into payroll</b> — each run auto-deducts the next instalment with the labelled line " <i>Welfare advance · Instalment N of M · {id} · {category}</i> ", and the welfare record auto-closes when fully repaid. Role gates: HR / Manager / Owner approve; Accountant / Owner / Manager disburse. Audit-logged under <b>HR</b> . See FR-PAY-2 / FR-WEL. Schema bump ( v21 ). |

| Version | Date       | Summary                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| 1.19    | 2026-05-24 | <p><b>POS start screen — service-type chooser; small plain-English pass.</b> The Front-of-House start screen now leads with <b>three equal tiles — Dine in · Takeout · Delivery</b> (Toast / Square / Lightspeed pattern). Picking <i>Dine in</i> expands into the table floor plan with a <i>Back to service type</i> link; <i>Takeout / Delivery</i> open the existing customer-capture modal directly. <b>Open tabs</b> sit above the chooser regardless of mode. Plain-English pass on the highest-friction jargon: rider P&amp;L now reads <b>Money in / Money out / Profit</b>; payslip statutory deductions are now <b>Income tax (PAYE 8%), Pension contribution (8%), Housing fund (NHF 2.5%)</b> under a "<i>what the law takes</i>" subheader; the welfare <i>repayable advance</i> checkbox is now plainly <b>"Pay it back from salary?"</b>. Self-audit acknowledging the remaining jargon (<i>Strong Room, A/R aging, Pour cost, Reorder point</i>) is listed in \$26.6 as a follow-up.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 1.20    | 2026-05-24 | <p><b>Nav grouping · ready-ticket notifications · financials · two-button-switcher cleanup · cashier RBAC.</b> (a) <b>Sidebar restructured into labelled sections</b> ( Operate · Inventory · People · Money · Grow · System ) for owner / manager / accountant / hr / storekeeper — Toast / Square / Lightspeed pattern, no flat 21-item list. Labels renamed for clarity (<i>Front of House on Shift</i> → <i>Shifts</i>, <i>Stock Transfers</i> → <i>Transfers</i>, <i>HR Dashboard</i> → <i>HR</i>, <i>Staff Welfare</i> → <i>Welfare</i>, <i>Inventory</i> → <i>Stock</i>, <i>Alerts &amp; Security</i> → <i>Alerts</i>, <i>Expenses</i> → <i>Petty cash</i> for cashier). (b) <b>Kitchen → waiter notification:</b> new <i>Ready to serve</i> strip on /cashier-home with each ready ticket + a <i>Served</i> button; same items surface in the global notification bell (Toast pattern). (c) <b>Financials</b> — new lib/financials.ts + /financials page (owner / manager / accountant) with <b>Profit &amp; Loss</b> (date-range) and <b>Balance Sheet</b> (snapshot). Every line leads with plain English ("<i>Money customers paid us</i>") and parenthesises the accounting term ("<i>Revenue</i>"). (d) <b>Three two-button mode-switchers eliminated:</b> AddRiderModal → split into HireInternalRiderModal + Add3PLPartnerModal (two separate buttons on /dispatch ); PostChargeModal → split into ChargeWalletModal + PostHouseChargeModal (two entries on the customer 360 gear menu, only the relevant one shown); DiscountModal % / ₺ toggle → <b>two side-by-side fields that cross-update</b> (edit either, both stay in sync). (e) <b>Cashier RBAC</b> tightened: /dispatch removed from cashier nav (they assign/complete deliveries from elsewhere); fleet-management actions (Hire rider, Add 3PL, Log expense, Settle COD) gated to owner/manager/storekeeper; Log fuel/repair hidden in the Rider 360 modal for non-fleet roles. (f) <b>POS split-payment</b> discoverability — header now reads "<i>Add another payment</i>" on second entry with "₺X still to collect" hint, clearer than the previous static "<i>Add Cash payment</i>" label. (g) /customers <b>cleaned up</b> — Complaints &amp; feedback and Marketing &amp; retention sections are now &lt;details&gt; blocks closed by default with status pills (e.g. "<i>2 open</i>" / "<i>5 lapsed</i>") so the page lands on the operational view (customers + AR).</p> |
| 1.21    | 2026-05-24 | <p><b>Industry-grade Financials · shift-close inventory count · login audit · inventory column reorder.</b> (a) <b>Financials dashboard rebuilt</b> for "run your business without an accountant" usability. Three big headline cards (<b>You made · You spent · You kept</b>) with Δ% vs prior period. Below them, <b>four restaurant health bars</b> (Food cost · Drink cost · Staff cost · Prime cost) coloured by NRA / Toast industry targets — green/amber/red bands and target ranges shown. Three tabs: <b>Profit &amp; Loss</b> with revenue split by stream (food / drink / other) and full operating breakdown; <b>Balance Sheet</b> with current vs fixed assets, liabilities, owner's equity; <b>Cash flow</b> in vs out. Every line leads with plain English and parenthesises the accounting term. (b) <b>Kitchen / Bar shift-close inventory count</b> — new ShiftCloseCountModal (Toast / Lightspeed / MarketMan pattern) listing the station's high-value items (Spirits / Wine / Protein / anything ≥ ₺2k cost) with per-row count inputs, live variance preview, <i>Show everything</i> toggle. Triggered by a <i>Closing count</i> button on /kitchen-home and /bar-home . Variance flags as over-pour / shrinkage in /alerts . (c) <b>Login audit trail</b> — new Security audit category. Every successful sign-in lands on /audit with the role + actor + branch + timestamp; failed PIN attempts also log (warning severity after 3 in a session) — PCI-DSS-style. (d) <b>Inventory table column reorder</b> — Item now leads (with the clickable category badge), SKU moved to penultimate column (it's a system ID, not what staff scan for). Matches Toast / Square / Lightspeed column order.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

| Version | Date       | Summary                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| 1.22    | 2026-05-24 | <b>Dispatch slim-down · vendor bulk-buy pricing · line filter on stock pickers.</b> (a) <code>/dispatch</code> <b>decluttered</b> . Active-delivery cards now lead with the address (the one thing dispatchers scan for) and bundle phone / COD / fee / time / rider into a muted footer. Rider grid replaced with a <b>compact list strip</b> — one row per rider with avatar, name, status dot, plate/phone, today's delivery count, COD-held chip with inline <i>Settle</i> , and clock-in/out + <i>Open</i> actions. P&L / fee revenue / ledger detail moved out of the index and into the Rider 360 modal where it belongs. (b) <b>Vendor bulk-buy pricing</b> (Toast / Lightspeed / MarketMan / xtraCHEF pattern). New <code>VendorPriceTier</code> + <code>VendorSkuPricing</code> types; pure-function <code>priceForQty(vendor, sku, qty)</code> resolves the active tier. The <b>PO modal</b> auto-fills unit cost from the applicable tier when SKU or quantity changes, shows a yellow "Add N more to drop to $\text{₦}X/\text{unit}$ " hint with savings preview, and footer-totals the <i>Bulk-buy savings</i> on the order. Seeded tiered pricing on Rice (₦2,400 → 2,200 → 2,000), Goat Meat, and Heineken so the flow is testable on first load. Manual cost edits still win for one-off negotiations. <b>Storekeeper-managed UI for editing tiers on <code>/vendors</code> is deferred to next iteration</b> — model + mutations are ready. Schema bump ( v22 ). (c) <b>Line filter on <code>StockRequestModal</code> and <code>NewTransferModal</code></b> — a <i>Section</i> chip strip (Kitchen / Bar / Juice Bar / Lounge), pre-set to the caller's station (bartender at <code>/bar-home</code> → opens with <b>Bar</b> selected; same for kitchen and juice bar). Only sections actually present in the source store are shown as chips. Strip hides entirely when only one section exists. |
| 1.23    | 2026-05-24 | <b>Always-on "Request from Strong Room" button on branch Main Store tabs.</b> Previously the only path from <code>/inventory</code> to the transfer-request flow was the amber <b>Suggest restock</b> banner, which only appeared when items had already dipped below par. That left a gap for <i>proactive</i> restocking — topping up before a busy weekend, or for the storekeeper who knows what they need without waiting for the system to flag it. The toolbar now carries a primary $\rightleftharpoons$ <b>Request from Strong Room</b> button on the Main Store tab of any branch (hidden at the Strong Room itself, hidden on sub-store tabs where the internal-stock-request button already covers it). Clicking it opens the existing <code>NewTransferModal</code> with no prefill — the operator picks their items from the full Strong Room catalogue (with the new <i>Section / Category</i> chip filters for fast narrowing). <i>New product</i> moves to a secondary-styled button next to it (admin-tier, not a daily task).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

**Note on accuracy.** This PRD is reverse-engineered from the working prototype. Where the prototype simulates a behaviour that production must implement properly (e.g. authentication, persistence), the requirement is written for the *target product* and the prototype's shortcut is called out in §15 (Current state) and §16 (Path to production).

## 2. Executive summary

NativeID ROS is a **multi-branch Restaurant Operating System** built for a Nigerian hospitality group operating several outlets in Lagos plus a central warehouse. It unifies — in one role-aware application — front-of-house order taking, kitchen and bar order routing, inventory and recipe costing, inter-branch stock transfers, procurement, petty cash, HR and payroll, delivery dispatch, customer relationship management, events/banqueting, analytics, and an immutable audit trail.

The product is designed around a **hub-and-spoke operating model**: a central **Strong Room** holds bulk stock and runs procurement; **branches** (Lekki, Ikoyi, Agungi) sell to customers and pull stock from the hub via tracked transfers. Every role sees a tailored workspace, every sensitive action is logged, and money/stock leakage is surfaced continuously.

The current build is a **high-fidelity functional prototype**: all behaviour is real (orders deduct stock, shifts reconcile, payroll computes), but data lives in the browser (React state + `localStorage`) rather than a

backend. The prototype is feature-complete enough to serve as the functional specification for a production build.

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## 3. Product vision & goals

### 3.1 Vision

Give an owner-operator of a growing restaurant group a **single, trustworthy system of record** that makes every branch run the same way, makes leakage visible the moment it happens, and lets the owner manage by exception from anywhere.

### 3.2 Goals

| #  | Goal                                | How the product serves it                                                                                                    |
|----|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| G1 | Eliminate revenue and stock leakage | Recipe-linked stock depletion, blind receiving, over-pour detection, shift reconciliation, waste logging, variance reporting |
| G2 | Standardise multi-branch operations | One app, identical workflows per role, central Strong Room governing supply                                                  |
| G3 | Enforce accountability              | PIN-gated overrides, manager approvals, immutable audit trail of every sensitive action                                      |
| G4 | Manage by exception                 | Role dashboards that surface only what needs a decision; live alerts; "needs your decision" queues                           |
| G5 | Run the full back office            | Procurement, petty cash, HR, payroll, fleet — not just front-of-house POS                                                    |
| G6 | Make data actionable                | Menu engineering, channel mix, peak-hour staffing, leakage %, branch comparison, morning brief                               |

### 3.3 Non-goals (current release)

- Not a public-facing ordering site or customer mobile app.
  - Not an accounting/GL system (it produces inputs for accounting, not statutory books).
  - Not a reservations/table-booking product for the public.
  - No hardware integration (printers, cash drawers, card terminals, scales) in the prototype.
- 

## 4. Background & problem statement

Independent and small-group restaurants in the Nigerian market typically run on a patchwork of a basic POS, spreadsheets, WhatsApp, and paper waybills. The consequences:

- **Stock leakage is invisible.** Bar over-pouring, kitchen waste, and theft are not measured until month-end stock-takes, if ever.
- **Branches drift.** Each outlet improvises its own process; the owner cannot compare like-for-like.
- **Cash is unaccountable.** Petty cash and till floats are reconciled informally; shortages are absorbed silently.



- **Procurement is opaque.** Supplier price creep, partial deliveries, and unpaid invoices are tracked in someone's head.
- **No audit trail.** Voids, discounts, and overrides leave no record, so abuse is undetectable.
- **The owner is a bottleneck.** Approvals happen by phone call; the owner has no remote, real-time view.

NativeID ROS addresses these by making the **operating process itself the software** — every transfer, count, void, expense, and payroll run is a structured, logged event scoped to a branch and an actor.

## 5. Users, roles & personas

The system defines **six staff roles**. Each role has a default landing route, a filtered navigation set, and a branch assignment. The **Owner** is the only role that can switch between branches; every other role is pinned to its home branch.

### 5.1 Role catalogue

| Role        | Persona                     | Home branch   | Default route      | Primary jobs                                                                           |
|-------------|-----------------------------|---------------|--------------------|----------------------------------------------------------------------------------------|
| owner       | Seun O. — Managing Director | All (roaming) | / Overview         | Group oversight, cross-branch comparison, high-value approvals, menu & events strategy |
| manager     | Tunde A. — Branch Manager   | Lekki         | /manager-dashboard | Run a branch, approve transfers/expenses, manage staff & payroll, monitor leakage      |
| cashier     | Ada O. — Cashier            | Lekki         | /cashier-home      | Take orders, run a till shift, handle deliveries & petty cash                          |
| kitchen     | Amara K. — Head Chef        | Lekki         | /kitchen-home      | Work the kitchen ticket queue                                                          |
| bartender   | Chukwu B. — Bartender       | Lekki         | /bar-home          | Work the bar ticket queue, run a bar shift, log waste                                  |
| storekeeper | Eze M. — Storekeeper        | Strong Room   | /store-home        | Manage central inventory, issue transfers, receive purchase orders                     |
| accountant  | Bola F. — Accountant        | All (roaming) | /expenses          | Approve petty cash, monitor expense leakage and accounts payable                       |
| hr          | Funke I. — HR Officer       | All (roaming) | /hr                | Run HR — onboarding, offboarding, scheduling, compliance documents, payroll oversight  |

### 5.2 Persona detail

**The Owner (Seun)** — wants to know, in 30 seconds, how every branch did today and what needs a decision. Spends time in Overview, Analytics, Audit Trail, Alerts. Approves discounts/expenses above branch limits. Roams branches via the branch switcher.

**The Branch Manager (Tunde)** — runs Lekki day-to-day. Lives in the Manager Dashboard "needs your decision" queue: approving stock transfers and petty cash, watching alerts, logging staff incidents, running payroll. Cannot see other branches' data.

**The Cashier (Ada)** — opens a till shift with a float, takes dine-in/takeout/delivery orders on the POS, charges via cash/card/transfer/split, and closes the shift by counting the drawer. Voids require a manager PIN.

**Kitchen & Bar staff (Amara, Chukwu)** — work a single screen: a live queue of tickets that they bump through New → Preparing → Ready → cleared. The bartender additionally runs a shift and logs waste.

**The Storekeeper (Eze)** — sits at the Strong Room. Receives supplier deliveries into central stock, approves and issues branch transfer requests, generates waybills.

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## 6. Scope

### 6.1 In scope (delivered in the prototype)

Authentication & role routing; six role workspaces; Front of House (order taking) with floor plan; Kitchen/Bar display; Inventory with batches & waste; Stock Transfers (hub-and-spoke); Purchase Orders & Vendors; Menu & Recipes; Expenses & Petty Cash; Staff/HR; Payroll; Dispatch & Fleet; CRM (customers, feedback, complaints); Events/BEO; Analytics; Audit Trail; Alerts; multi-branch model; CSV import/export.

### 6.2 Out of scope (this release)

Real backend/database; real authentication & secrets management; hardware peripherals; payment-gateway integration; offline mode; native mobile apps; multi-currency; multi-tenant (multiple restaurant groups); statutory accounting exports; public ordering.

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## 7. System context & architecture

### 7.1 Technology stack

| Layer         | Choice                                                                                          |
|---------------|-------------------------------------------------------------------------------------------------|
| Framework     | Next.js 15.5 (App Router)                                                                       |
| UI runtime    | React 19                                                                                        |
| Language      | TypeScript 5.8                                                                                  |
| Styling       | Tailwind CSS v4 (configured in <code>app/globals.css</code> , <code>oklch</code> token palette) |
| Component kit | shadcn/ui (new-york style), Radix primitives, lucide-react icons                                |
| Charts        | Recharts (available); most charts hand-rolled with divs                                         |
| Toasts        | Sonner                                                                                          |
| Font          | Outfit (Google Fonts)                                                                           |
| Hosting       | Vercel (framework preset must be <b>Next.js</b> ; see <code>vercel.json</code> )                |



## 7.2 Application architecture

- `AuthProvider` (`lib/auth.tsx`) — client-side session. Holds the current `StaffUser`, exposes `login / logout`. Session persisted in `localStorage` key `nativeid_user`.
- `StoreProvider` (`lib/store.tsx`) — the entire domain model and business logic in one React context: ~30 entity collections plus ~60 mutation functions. State persisted in `localStorage` key `haven11_store_v11`, seeded with realistic demo data on first load.
- `AuthGuard` — wraps the app; redirects unauthenticated users to `/login`, shows a splash while resolving.
- `AppShell` — every page renders inside it: role-filtered sidebar, sticky header, branch switcher, notification bell. Also exports the `PageSection` (card) and `Stat` (KPI tile) primitives.
- **Pages** — one route per module under `app/`. Each page reads from `useStore()` / `useAuth()` and calls store mutations.

## 7.3 The data seam

All data and logic currently sit in `lib/store.tsx`. **The production migration path** is to replace the in-context state and mutation bodies with API calls to a real backend, keeping the `useStore()` hook surface (the ~60 named functions) stable so pages do not change. This is the single most important architectural seam.

## 8. Information architecture & navigation

### 8.1 Route → role access matrix

Navigation is filtered per role by `roleNav` in `AppShell.tsx`. ✓ = module appears in that role's sidebar.

| Route                           | Module                | owner | manager | cashier | kitchen | bartender | storekeeper |
|---------------------------------|-----------------------|-------|---------|---------|---------|-----------|-------------|
| <code>/</code>                  | Operations Overview   | ✓     |         |         |         |           |             |
| <code>/manager-dashboard</code> | Branch Dashboard      |       | ✓       |         |         |           |             |
| <code>/cashier-home</code>      | Cashier — My Shift    |       |         | ✓       |         |           |             |
| <code>/kitchen-home</code>      | Kitchen Display       |       |         |         | ✓       |           |             |
| <code>/bar-home</code>          | Bar Queue             |       |         |         |         | ✓         |             |
| <code>/store-home</code>        | Store Home            |       |         |         |         |           | ✓           |
| <code>/pos</code>               | Front of House        | ✓     | ✓       | ✓       |         | ✓         |             |
| <code>/kitchen-bar</code>       | Kitchen & Bar Display | ✓     | ✓       |         |         |           |             |
| <code>/inventory</code>         | Inventory             | ✓     | ✓       |         |         |           | ✓           |
| <code>/transfers</code>         | Stock Transfers       | ✓     | ✓       |         |         |           | ✓           |
| <code>/purchase-orders</code>   | Procurement           | ✓     | ✓       |         |         |           | ✓           |
| <code>/vendors</code>           | Vendors               | ✓     | ✓       |         |         |           | ✓           |
| <code>/menu</code>              | Menu & Recipes        | ✓     |         |         |         |           |             |

| Route                   | Module                  | owner | manager | cashier | kitchen | bartender | storekeeper |
|-------------------------|-------------------------|-------|---------|---------|---------|-----------|-------------|
| <code>/dispatch</code>  | Dispatch & Fleet        | ✓     | ✓       | ✓       |         |           |             |
| <code>/cashier</code>   | Front of House on Shift | ✓     | ✓       |         |         |           |             |
| <code>/expenses</code>  | Expenses & Petty Cash   | ✓     | ✓       | ✓       |         |           | ✓           |
| <code>/staff</code>     | Staff (HR)              | ✓     | ✓       |         |         |           |             |
| <code>/hr</code>        | HR Dashboard            | ✓     | ✓       |         |         |           |             |
| <code>/payroll</code>   | Payroll                 | ✓     | ✓       |         |         |           |             |
| <code>/events</code>    | Events / BEO            | ✓     |         |         |         |           |             |
| <code>/customers</code> | Customers (CRM)         | ✓     | ✓       |         |         |           |             |
| <code>/reports</code>   | Analytics               | ✓     | ✓       |         |         |           |             |
| <code>/audit</code>     | Audit Trail             | ✓     | ✓       |         |         |           |             |
| <code>/alerts</code>    | Alerts & Security       | ✓     |         |         |         |           |             |

The **Accountant** (added in v1.3) is a roaming finance role — it sees **Expenses, Procurement, Vendors** and the **Audit Trail**. The **HR Officer** (added in v1.5) is a roaming HR role — sees **HR Dashboard, Staff, Payroll** and the **Audit Trail**.

**Known gap (security):** AuthGuard only enforces *authenticated vs not*. It does **not** hard-gate routes by role — the matrix above is enforced only by hiding sidebar links. A signed-in user who types another route directly is not blocked. Production must enforce this matrix server-side. See FR-AUTH-5 and §17.

## 8.2 Global shell elements

- **Sidebar** — branded header, role-filtered nav, current-user card, sign out. Collapses to a drawer on mobile.
- **Header** — page title/subtitle, **branch switcher**, **notification bell**.
- **Branch switcher** — owner sees a dropdown of all branches; all other roles see a static chip of their home branch.
- **Notification bell** — live count + dropdown derived from the current branch's state: over-pour incidents, expiring batches ( $\leq 14$  days), and low/out-of-stock items. Items deep-link to `/alerts` or `/inventory`.

## 9. Domain model

The store defines the following core entities (see `lib/store.tsx` for full field lists).

| Entity | Purpose / key fields                                                                                                                                      |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Branch | <code>id</code> , <code>name</code> , <code>kind</code> ( <code>hub</code>   <code>branch</code> ). Four seeded: Strong Room (hub), Lekki, Ikoyi, Agungi. |

| Entity                                | Purpose / key fields                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| InventoryItem                         | Stock row keyed by branch + <b>location</b> + SKU: sku, branch, location (store / kitchen / bar / juice-bar), onHand, reorder (par level), unit, cost, line. Optional altUnit + altOnHand capture a second measurement (e.g. 30 kg yam recorded alongside 10 pcs). One shared product catalogue — no duplicate SKUs across locations.                                                               |
| MenuItem                              | name, category, price, emoji, status (Available/Sold out/Coming soon), recipe (SKU + qty lines).                                                                                                                                                                                                                                                                                                    |
| Order                                 | Sale: branch, table, channel, customer, lines, subtotal, discount, vat (7.5%), total, payments, splitWays, staffName, shiftId, status ( <b>On hold / Closed</b> ), voided. An order is Closed only once payment is received.                                                                                                                                                                        |
| Shift                                 | Till/bar session: staffId, role, branch, openingFloat, countedCash, openedAt / closedAt, status.                                                                                                                                                                                                                                                                                                    |
| Ticket                                | Kitchen/Bar work item: station, label, items, status (New/Preparing/Ready).                                                                                                                                                                                                                                                                                                                         |
| TableRec                              | Floor plan unit: label, zone, seats, status, guests, orderTotal.                                                                                                                                                                                                                                                                                                                                    |
| Transfer                              | Inter-branch waybill: fromBranch (always hub), toBranch, lines (requested/issued/received qty), status, valueAtCost. An issued transfer carries a printable receipt.                                                                                                                                                                                                                                |
| StockRequest                          | Internal request to move stock from a branch Main Store into a Kitchen / Bar / Juice Bar sub-store: toLocation, lines, status (Requested/Issued).                                                                                                                                                                                                                                                   |
| Vendor                                | Supplier: name, contact, terms (COD/Net 15/Net 30), tin, category.                                                                                                                                                                                                                                                                                                                                  |
| PurchaseOrder                         | Procurement: vendorId, branch (destination), lines, status ( <b>Pending Approval</b> → Ordered → Partially/Received, or Rejected), expectedDate, total, paid.                                                                                                                                                                                                                                       |
| Batch                                 | Perishable lot with expiry date, for FIFO/expiry tracking.                                                                                                                                                                                                                                                                                                                                          |
| PriceChange                           | Logged supplier cost variance (old → new).                                                                                                                                                                                                                                                                                                                                                          |
| WasteEntry / StockCount               | Inventory loss & variance records (count carries an overPour flag). WasteEntry may include an attached <b>photo</b> (filename + data URL for small files) captured on the prep floor.                                                                                                                                                                                                               |
| ExpenseRequest / Wallet               | Petty cash requisition lifecycle + per-branch imprest wallet.                                                                                                                                                                                                                                                                                                                                       |
| Employee / Attendance / Disciplinary  | HR records, clock-ins with lateness, incidents & commendations. Employee.status is <b>Active / Suspended / Offboarded</b> (offboarding never deletes); Employee.weeklySchedule (Mon–Sun) drives the roster. Employee.compliance holds three uploaded ComplianceDocs — guarantor form, signed contract, ID card — each with filename, size, upload date and a downloadable data URL for small files. |
| PayrollRun / Payslip                  | Monthly payroll with statutory deductions.                                                                                                                                                                                                                                                                                                                                                          |
| Rider / DeliveryJob                   | Fleet members and delivery jobs (COD tracking).                                                                                                                                                                                                                                                                                                                                                     |
| Customer / Feedback / ComplaintTicket | CRM golden record (keyed by phone), ratings, complaints.                                                                                                                                                                                                                                                                                                                                            |

| Entity          | Purpose / key fields                                               |
|-----------------|--------------------------------------------------------------------|
| RestaurantEvent | Banquet/event booking with cost lines, deposit, P&L.               |
| AuditEntry      | Immutable who/what/when/where/amount record; category , severity . |

## 9.1 Reference data & constants

- **Branches:** Strong Room ( strong-room , hub), Lekki, Ikoyi, Agungi.
- **Inventory locations:** each branch holds stock in a **Main Store** plus **Kitchen**, **Bar** and **Juice Bar** sub-stores; the Strong Room has a single store. line (Kitchen/Bar/Lounge/Juice Bar) categorises a product; location is where the stock physically sits.
- **VAT:** 7.5% on (subtotal – discount).
- **Payroll statutory rates:** PAYE 8% of gross; Pension 8% of base; NHF 2.5% of base; lateness ₦500 per late mark (≥15 min).
- **Currency:** Nigerian Naira (₦). All amounts integer Naira.

## 10. Cross-cutting requirements

These behaviours apply across all modules.

### 10.1 Multi-branch operating model

- **FR-BR-1** Every stock row, order, shift, transfer, expense, employee, and audit entry is scoped to a branch .
- **FR-BR-2** The **Owner** may switch the operating branch via the header switcher; all data re-scopes immediately.
- **FR-BR-3** All non-owner roles are **pinned** to their home branch; AppShell forces currentBranch back to the user's branch on every render.
- **FR-BR-4** The **Strong Room** is a hub: it holds bulk stock and is the sole fromBranch of every transfer. Branches receive stock only via transfers from the hub or via direct PO receipt.

### 10.2 Role-based access control (RBAC)

- **FR-RBAC-1** Sidebar navigation is filtered to the role's roleNav set.
- **FR-RBAC-2** Owner sees group-wide data on shared modules (Audit, Reports, Cashier); manager and below see only their branch.
- **FR-RBAC-3** (*Production*) Route access must be enforced server-side per the §8.1 matrix, not only by hiding links.

### 10.3 Immutable audit trail

- **FR-AUD-1** The store automatically writes an AuditEntry for every sensitive action: order void, discount applied, shift close, stock variance/over-pour, waste, transfer approve/reject/issue/receive, goods received, cost-price change, expense approve/reject/disburse/reconcile, wallet top-up, disciplinary, payroll run.
- **FR-AUD-2** Each entry records actor , branch , category , action , human-readable detail , optional ref and amount , and severity (info/warning).
- **FR-AUD-3** Entries cannot be edited or deleted from the UI.

## 10.4 Approvals & PIN gating

- **FR-APR-1** A `ManagerApprovalModal` validates a 4-digit PIN against the staff roster; an `ownerOnly` flag restricts to the owner.
- **FR-APR-2** POS discounts above ₦500 require a manager/owner PIN.
- **FR-APR-3** Order voids require manager approval and a reason.
- **FR-APR-4** Petty cash approval is tiered by amount (see §11.13).

## 10.5 Notifications & alerts

- **FR-NOT-1** The header bell shows a live unread count derived from current-branch state (over-pours, expiring batches, low/out stock).
- **FR-NOT-2** Alerts deep-link to the relevant module; "mark all read" clears the count (session-local).

## 10.6 Data import / export

- **FR-IO-1** Inventory and Menu support **bulk CSV import** (upload or paste) via `ImportModal`, with a downloadable template, tolerant column matching, and a live preview. Existing rows update; new rows insert.
- **FR-IO-2** Inventory, Analytics (menu engineering), Payroll (bank schedule), and Audit Trail support **CSV export** of the current filtered view.

## 10.7 Persistence

- **FR-PER-1** (*Prototype*) All state persists to `localStorage` and survives refresh; a `resetAll` restores seed data.
- **FR-PER-2** (*Production*) State must persist to a backend database with concurrent multi-device access.

## 10.8 Design system

- **FR-DS-1** All pages use `AppShell` + `PageSection` + `Stat` for layout consistency.
- **FR-DS-2** Status colours are semantic and consistent: green = OK/positive, amber `text-warning` = caution/low, red `destructive` = critical/loss, deep-green `primary` = active/CTA.
- **FR-DS-3** Fully responsive; sidebar becomes a drawer below `lg`.

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# 11. Functional requirements by module

Each module below lists its purpose, primary users, and numbered functional requirements.

## 11.1 Authentication & login ( `/login` )

**Users:** all. **Purpose:** identify the staff member and route them to their workspace.

- **FR-AUTH-1** The login screen shows the staff roster as selectable avatar cards, each with a role-coloured icon ("Who's working today?").
- **FR-AUTH-2** Selecting a staff member opens a 4-digit PIN pad; the PIN auto-submits on the 4th digit.
- **FR-AUTH-3** A correct PIN starts a session and redirects to the role's `defaultRoute`; a wrong PIN shows a shake animation and error, and clears the entry.

- **FR-AUTH-4** A session persists across refreshes; an already-authenticated user is redirected away from `/login`.
- **FR-AUTH-5** (*Production*) Replace PINs with proper credentials/secrets, hash-verified server-side; enforce route-level RBAC; support session expiry and idle auto-lock (the Alerts page already advertises "idle sessions auto-locked").

## 11.2 Operations Overview — Owner dashboard ( / )

**Users:** owner. **Purpose:** 30-second read on a branch's live trading.

- **FR-OVW-1** KPI tiles: **Sales today** (¥ + order count), **Covers seated**, **Open tickets** (kitchen+bar), **Stock alerts** (amber when > 0).
- **FR-OVW-2 Revenue by channel** panel — Dine-in / Takeout / Delivery split with amounts, %, and bars.
- **FR-OVW-3 Live alerts** panel — top over-pours and low-stock items; each deep-links to `/alerts`.
- **FR-OVW-4 Recent orders** table — last 6 orders with channel, item count, total, age.
- **FR-OVW-5 Inventory health** — per line (Kitchen/Bar/Lounge) a stacked OK/Low/Out bar with counts.
- **FR-OVW-6** All figures exclude voided orders and re-scope when the owner switches branch.

## 11.3 Branch Dashboard — Manager ( /manager-dashboard )

**Users:** manager. **Purpose:** run a branch and clear the decision queue.

- **FR-MGR-1** KPI tiles: Sales today, Open tickets, **Pending approvals** (transfers + expenses, amber when > 0), Stock alerts.
- **FR-MGR-2 Needs your decision** — combined queue of pending transfers and pending expenses; each item deep-links to its approval page.
- **FR-MGR-3 Live alerts** panel (over-pours + low stock).
- **FR-MGR-4 Recent orders** list (last 6).
- **FR-MGR-5 Quick links** to POS, Stock Transfers, Inventory, Expenses.

## 11.4 Role home screens

Lightweight landing pages for operational roles.

- **FR-HOME-1 — Cashier home ( /cashier-home )**: shift banner; large "New Order" CTA → `/pos`; shift KPIs (Cash/Card/Transfer totals + order count); floor plan with seat/clear-table actions; recent orders list with **void** (manager-approved).
- **FR-HOME-2 — Kitchen home ( /kitchen-home )**: status counts (Waiting/Cooking/Ready); large ticket cards; each card advances New → Preparing → Ready → cleared; tickets ≥12 min flagged overdue.
- **FR-HOME-3 — Bar home ( /bar-home )**: shift banner; "Record waste" action; status counts (Waiting/Making/Ready); ticket cards with the same advance flow; tickets ≥8 min flagged slow.
- **FR-HOME-4 — Store home ( /store-home )**: task aggregator — low stock, transfers to issue/receive, POs to receive, batches expiring ≤14 days; a count of "tasks that need you today"; CTA to raise a transfer.

## 11.5 Front of House ( /pos )

**Users:** owner, manager, cashier, bartender. **Purpose:** take orders, run open tabs, and take payment.



**Renamed in v1.1.** Formerly "Point of Sale (POS)" — renamed to **Front of House (FOH)** because "POS" is widely read as the physical card/payment terminal, which confused users. The order-taking screen is labelled Front of House; the route stays `/pos`.

- **FR-POS-1 — Service-type chooser:** the start screen presents **three equal tiles** — **Dine in** · **Takeout** · **Delivery** (Toast / Square / Lightspeed pattern). Picking **Dine in** expands into the **zoned floor plan** (Indoor / Terrace / Bar) with a *Back to service type* link; tapping any available table starts a seating flow (occupied / reserved tables are not selectable). Picking **Takeout** or **Delivery** opens the customer-capture modal directly. The Dine-in tile shows live `available` · `seated` counts to scan branch capacity at a glance without expanding the floor plan. **Open tabs** sit above the chooser regardless of mode — they're always actionable.
- **FR-POS-2 — Seat guests:** seating a table captures guest count (bounded by table capacity) and marks the table occupied.
- **FR-POS-3 — Service forms:** Takeout captures customer name, phone, pickup time (ASAP / 15 min / 30 min / 1 hr); Delivery captures name, phone, address, and a delivery fee (default ₦1,500).
- **FR-POS-4 — Menu & cart:** category-tabbed menu grid (Available items only); tapping an item adds/increments a cart line; per-line quantity  $\pm$  and delete; sticky cart sidebar shows order label and context.
- **FR-POS-5 — Modifiers:** each line can take modifiers (Extra cheese +₦300, Extra sauce +₦200, Extra protein +₦800, Extra spicy, No onions, Well done) and a free-text instruction; both print on the kitchen/bar ticket.
- **FR-POS-6 — Discounts:** apply a percentage or fixed-amount discount with a reason; discounts over ₦500 require a manager/owner PIN; the discount is written to the audit trail.
- **FR-POS-7 — Totals:** cart shows Subtotal, Discount, VAT 7.5%, Total.
- **FR-POS-8 — Payment:** split the bill 1–8 ways; take one or more payments across **Cash / Card / Transfer**; quick-fill "remaining" and "one share"; show paid / still-due / change.
- **FR-POS-9 — Cash on delivery:** delivery orders may complete as COD (rider collects total + fee).
- **FR-POS-10 — Completing a sale** ( `recordSale` ) atomically: creates the `Order` ; deducts each recipe ingredient from **this branch's** stock; raises Kitchen and/or Bar tickets (drinks/cocktails route to Bar); for delivery, raises a `DeliveryJob` ; for takeout/delivery, upserts the customer golden record by phone; links the order to the cashier's open shift; logs a discount audit entry if applicable.
- **FR-POS-11 — Confirmation:** a done screen shows order number, payment breakdown, change, and customer details; "New order" resets.
- **FR-POS-12 — Open a tab:** an **Open tab** action places the order as **On hold** — items fire to the kitchen/bar and stock is deducted immediately, with no payment taken. The order joins the **"Open tabs"** list on the Front of House start screen (seated and delivery alike).
- **FR-POS-13 — Amend an open tab:** tapping an open tab resumes it — items already sent to the kitchen are shown locked; the cashier adds more items and either **sends the new items to the kitchen** (the tab stays open) or proceeds to close it. Newly-added items fire fresh kitchen/bar tickets and deduct their stock; the tab total recomputes.
- **FR-POS-14 — Close & pay:** closing a tab takes payment (cash or POS/card terminal); the order becomes **Closed**, a dine-in table is freed, and only then does it count as a sale. On-hold tabs are excluded from revenue on dashboards and analytics.

## 11.6 Kitchen & Bar Display ( /kitchen-bar , plus role homes)

**Users:** owner, manager (combined view); kitchen/bartender (single-station homes).

- **FR-KDS-1** Tickets are generated by Front of House and routed by station — drinks & cocktails to **Bar**, everything else to **Kitchen**.
- **FR-KDS-2** Each ticket shows order ref, table/channel label, item lines with modifiers/notes, status, and age.
- **FR-KDS-3** Staff advance a ticket New → Preparing → Ready; "serve & clear" removes a Ready ticket; "Mark ready" jumps straight to Ready.
- **FR-KDS-4** Ticket age is colour-flagged (kitchen overdue ≥ 12 min; bar slow ≥ 8 min).
- **FR-KDS-5** Voiding the source order removes its tickets.

## 11.7 Front of House on Shift ( /cashier ) + shift lifecycle

**Users:** owner, manager (ledger view); cashier/waiter/bartender (own shift via `ShiftBanner` ).

**Renamed in v1.1.** Formerly "Cashier & Shifts" — renamed to **Front of House on Shift** because the module serves **waiters as well as cashiers**: anyone who works an opening balance and handles cash.

- **FR-SHF-1** A cashier/bartender opens a shift with an **opening cash float** (default ₦50,000).
- **FR-SHF-2** Orders rung during a shift link to it; the shift banner shows elapsed time, order count, sales, and waste.
- **FR-SHF-3** Closing a shift captures **counted cash**; the system computes variance = counted – (float + sales) and writes a "Shift closed" audit entry (warning if short).
- **FR-SHF-4** A **bartender** closing a shift additionally performs a **bar stock count** per Bar SKU; negative variances are flagged as **over-pours** attributed to the bartender.
- **FR-SHF-5** The `/cashier` page is the **live reconciliation ledger**: per shift — staff member, date, opened time, shift period, counted cash, variance (colour-coded), status; KPIs for cash in tills, reconciled variance, open shifts. Branch-scoped (owner sees all).
- **FR-SHF-6 — Shift history:** the ledger shows each shift's **date** and offers a **date drop-down** to review past days, not just today.
- **FR-SHF-7 — Configurable shift periods:** on clock-in, staff pick a **shift period** — **Full day** (single-shift branch) or **1st / 2nd / 3rd shift** (multi-shift branch); the period shows on the ledger.

## 11.8 Inventory ( /inventory )

**Users:** owner, manager, storekeeper. **Purpose:** branch stock control.

- **FR-INV-1** KPI tiles: Total SKUs, Low stock (click to filter), Out of stock (click to filter), Stock value ( $\Sigma$  onHand × cost).
- **FR-INV-2** Searchable, filterable (by line, by status) inventory table; status derived as OK / Low ( $\leq$  reorder) / Out ( $\leq 0$ ).
- **FR-INV-3 New SKU** — add an item with auto-generated SKU (line-prefixed), unit, on-hand, reorder level, cost.
- **FR-INV-4 Adjust** — per item, either **Receive stock** (add quantity) or **Stock count** (set actual; logs a `StockCount` with variance and cost; Bar negative variance = over-pour, attributed to the bartender on shift if any).

- **FR-INV-5 Waste** — log spoilage/loss via `WasteModal` (item, quantity, reason); deducts stock, records cost, attributes to staff/shift.
- **FR-INV-6 Activity log** (toggle) — recent stock-count variances and waste entries.
- **FR-INV-7 Batch & expiry register** — perishable batches sorted soonest-expiry-first, colour-coded ( $\leq 7$  days red,  $\leq 14$  amber); supports FIFO discipline.
- **FR-INV-8 Import** (bulk CSV upsert) and **Export** (filtered CSV) of inventory.
- **FR-INV-9** All inventory data is scoped to the current branch; Front-of-House sales deduct stock here in real time.
- **FR-INV-10 — Section separation:** inventory carries a **section** — **Kitchen, Bar, Juice Bar**, or Lounge — so each section is filtered, audited, and managed independently, reflecting that sections deplete stock at different rates.
- **FR-INV-11 — Stock locations:** each branch keeps stock in a **Main Store** plus **Kitchen, Bar** and **Juice Bar** sub-stores. The Inventory page has a location tab per store; counts, waste, receiving and adjustments act on the selected location. A product is defined once (one shared catalogue) — there are no duplicate goods across the Main Store and the sub-stores.
- **FR-INV-12 — Internal stock requests:** the Kitchen, Bar and Juice Bar **request stock from the branch Main Store**; the request is **issued** from the Main Store, moving stock into the requesting sub-store. Front-of-House sales then deduct each recipe ingredient from the sub-store matching its line. The request modal is a **multi-select picker** — the operator sees the Main Store as a single scrollable list grouped by category, with a search box, category chips, a "tick all visible" bulk action, and a per-row checkbox + qty input. Built for 50+-item requests in one screen, not row-by-row.
- **FR-INV-13 — Categories:** every product carries a **category** (Protein, Produce, Grains, Spices, Oils & Fats, Dairy, Beer, Spirits, Wine, Mixers, Hot Drinks, Soft Drinks, Cleaning, Other). The Inventory list shows the category as a clickable badge in the Item cell (click to filter), exposes Category as a chip filter alongside Line and Status, and the free-text search matches name, SKU **and** category — so typing "spice" surfaces all spices, "protein" surfaces all proteins. The stock-request picker groups by the same category vocabulary. CSV import/export carries a `Category` column (an unknown value coerces to "Other" rather than crashing the import).
- **FR-INV-14 — Category management (owner/manager):** the category vocabulary lives in `state.inventoryCategories`, not as a hardcoded constant, and is curated through a focused **Manage categories** modal — add, rename, delete in one list with per-category product counts. **Rename cascades** to every stock row using that category. **Delete is blocked** while a category is in use (counter > 0); the operator must reassign products to another category first. The modal is reached from **two unobtrusive entry points** — a `Manage categories` → link in the inventory Filter popover footer, and a `Manage...` link beside the Category dropdown in the **New product** dialog (the Linear/Notion pattern: no new nav item, manage from where you'd create). Cashier, kitchen, bartender, storekeeper, accountant and HR see the categories but cannot edit them. Seeded with the 14 defaults from FR-INV-13.
- **FR-RPL — Suggested restock (manager-in-the-loop auto-replenishment):**
  - **Trigger:** any item with `onHand  $\leq$  reorder` (the par level) at the active location.
  - **Suggested quantity:** `target - onHand` where `target = reorder  $\times$  2` — the Toast / National Restaurant Association rule of thumb that stocks up to 2 $\times$  par so the next cycle has buffer. Rounded up to a sensible step (1 for discrete units like pcs/btl/bunch; 0.1 for continuous units like kg/L). Capped at source availability — never suggest more than the source actually has on hand.
  - **Source of restock:** a branch Main Store restocks from the Strong Room (via `/transfers` waybill flow). A sub-store (Kitchen / Bar / Juice Bar) restocks from its own branch Main Store (via the internal stock-request flow).

- **UX:** when low items exist at the active location, an amber banner on the inventory page reads "*N items running low — restock from {source}?*" and lists the first few. Clicking opens the existing multi-select picker (FR-INV-12 / FR-TRF-2) with the low items already ticked and quantities pre-filled. The operator can edit anything before submitting; submission goes through the normal approval flow (FR-TRF-3 management approval for hub→branch; the Main Store storekeeper issues for sub-stores).
- **Industry positioning:** *suggest, never auto-submit*. Restaurant inventory has too much context (events, weather, holidays) for unattended ordering — manager-in-the-loop is the universal pattern across Toast, Lightspeed, Square for Restaurants, and Olo. The system does the math; the human does the judgement.
- **Roles:** the banner is visible to anyone who can see the inventory page (owner / manager / storekeeper). Submission then follows the same role gates as a manual request.
- **Implementation:** pure helper `suggestedRestockQty(item)`; the shared `NewTransferModal` and `StockRequestModal` both accept a `prefill: { sku, qty }[]` prop that pre-ticks the picker.

## 11.9 Stock Transfers ( /transfers )

**Users:** owner, manager, storekeeper. **Purpose:** govern hub→branch stock movement.

- **FR-TRF-1** KPI tiles: Awaiting approval, Approved, In transit, Disputed.
- **FR-TRF-2 Request** — a branch raises a transfer request from the Strong Room with a reason (Low stock / Event prep / Spoilage replacement / Emergency) and line items. The request modal is a **multi-select checklist** matching the internal stock-request picker (FR-INV-12): a single scrollable list of every item in the Strong Room's Main Store grouped by category with sticky headers, a search box (matches name / SKU / category), a category-chip filter, **"Tick all visible"** bulk action, and per-row checkbox + qty input. Each row shows live Strong Room availability so the requester knows what's actually in stock. Footer shows "*X/Y ready · Z total units*" live as items are ticked. Built for 50-item requests in one screen, not row-by-row. (Disabled while the operating context is the hub.)
- **FR-TRF-3** Lifecycle: **Requested** → **Approved/Rejected** → **Issued** → **Received** (or **Disputed**).
- **FR-TRF-4 Issue** — the Strong Room confirms dispatched quantities (default = requested), generates a waybill, deducts hub stock, and bills the destination branch at cost ( `valueAtCost` ).
- **FR-TRF-5 Receive** — the destination counts what physically arrived; any shortfall vs issued quantity flags the transfer **Disputed** (a goods-in-transit loss escalated for investigation); received stock is added to the destination branch.
- **FR-TRF-6 — Actions are context-aware (source vs destination):** approve / reject / issue are **source-side actions** — they only render when the viewer is currently in the source branch (the Strong Room for branch requests). A branch viewing its own request sees an italic status hint instead of action buttons ("*Awaiting Strong Room approval*" → "*Approved · Strong Room preparing*" → "*In transit · awaiting receipt*"). Receive is the **destination-side action** and the source instead sees "*In transit · awaiting receipt*" while waiting. The Receipt waybill view is available to both sides on Issued / Received / Disputed. This prevents a branch from rejecting (or approving, or issuing) its own request.
- **FR-TRF-6** Every state change writes an audit entry. Non-owners see only transfers involving their branch.
- **FR-TRF-7 — Issue receipt:** the Strong Room sells stock to branches — an issued transfer produces a printable **receipt** (waybill ref, destination, line items with unit cost and totals, grand total at cost).

## 11.10 Procurement ( /purchase-orders )

**Users:** owner, manager, storekeeper (finance sections owner/manager only).

**Renamed in v1.1.** The module is labelled **Procurement** (formerly "Purchase Orders" / "process order") — clearer for staff whose task is to *order inventory* for the kitchen and bar. A Purchase Order is the document Procurement produces.

- **FR-PO-1** KPI tiles: Open POs, Awaiting delivery; finance-only: Unpaid invoices, Accounts payable total.
- **FR-PO-2 Create PO** — select a vendor, expected date, and line items (SKU, quantity, unit cost auto-prefilled from catalogue); delivered to the Strong Room.
- **FR-PO-3 Receive goods** — count received quantities with optional **blind receiving** (hides ordered quantities to force honest counts); capture per-line actual unit cost and optional expiry date.
- **FR-PO-4** On receipt: stock is added to the hub; a cost variance vs catalogue creates a **PriceChange** and rolls the cost forward (audit-logged); an expiry date creates a **Batch**; status becomes Partially Received or Received.
- **FR-PO-5 Mark paid** — record invoice payment; status badge updates.
- **FR-PO-6 Accounts payable — aging** — outstanding POs by age (red after 30 days).
- **FR-PO-7 Supplier price changes** — log of received-cost variances with % change.
- **FR-PO-8 — Management approval:** a purchase order is created as **Pending Approval** and must be **approved by management** (owner/manager) before it becomes **Ordered** (sent to the vendor); it can also be **Rejected**. Goods can only be received once approved.
- **FR-PO-9 — Destination is the current branch (no cross-branch requesting):** a PO always delivers to the branch the user is currently viewing — the **Deliver to** field is a read-only label, not a dropdown. To raise a PO for another branch, the user switches branches via the header BranchSwitcher (owners, managers, accountants and HR can roam; branch-scoped roles only ever see their own branch). Combined with **FR-TR-2** (which already pins `toBranch = currentBranch` on new transfers), this means **branches cannot request products to be delivered to other branches** through any path in the app.

### 11.11 Vendors ( /vendors )

**Users:** owner, manager, storekeeper.

- **FR-VND-1** KPI tiles: Vendors, Outstanding payables, Open POs.
- **FR-VND-2** Vendor database table: name, category, payment terms, contact, amount owed.
- **FR-VND-3 New vendor** — name, contact, phone, email, TIN, terms (COD/Net 7/15/30), category.
- **FR-VND-4 Vendor detail** — contact card, lifetime spend, outstanding balance, and full purchase-order history.

### 11.12 Menu & Recipes ( /menu )

**Users:** owner.

- **FR-MNU-1** Category-filtered grid of menu items; each card shows price, recipe cost, **profit margin %** (red < 15%), ingredient count, and status.
- **FR-MNU-2 New menu item** — name, category, price, emoji; immediately orderable in the POS.
- **FR-MNU-3 Edit item** — name, price, availability, and a **recipe builder** linking inventory SKUs + quantities; a live summary shows recipe cost vs price vs margin.
- **FR-MNU-4 View recipe** — per-ingredient cost breakdown; recipes are what drive real-time stock depletion on every sale.
- **FR-MNU-5 Bulk import** menu items via CSV (recipes linked afterwards by editing).

## 11.13 Expenses & Petty Cash ( /expenses )

**Users:** owner, manager, cashier, storekeeper.

- **FR-EXP-1** KPI tiles: Pending approval, Awaiting reconciliation, Spent this period.
- **FR-EXP-2 Branch wallet** — per-branch imprest wallet with balance vs target float; low-balance warning; **request top-up** (reimbursement from Head Office).
- **FR-EXP-3 New request** — category (8 categories, each with a monthly budget), amount, description; the system shows remaining category budget and the approval tier.
- **FR-EXP-4 Accountant approval:** petty-cash requests are approved (or rejected) by the **Accountant** — a dedicated role with its own login and finance workspace. The Owner may also approve.
- **FR-EXP-5 Lifecycle:** **Pending** → **Approved/Rejected** → **Disbursed** → **Reconciled**. Disbursement debits the branch wallet.
- **FR-EXP-6 Reconcile** — record actual spend and receipt (or a manager waiver); unspent change is returned to the wallet.
- **FR-EXP-7** A staff member with an open (disbursed, unreconciled) requisition is blocked from raising a new one.
- **FR-EXP-8 Expense leakage** report (manager/owner) — spend vs budget per category, over-budget bars in red.
- **FR-EXP-9** Approve/reject/disburse/reconcile and top-up all write audit entries.

## 11.14 Staff / HR ( /staff )

**Users:** owner, manager.

- **FR-HR-1** KPI tiles: Active staff, Clocked in today, Compliance gaps, Certs expiring ( $\leq 30$  days).
- **FR-HR-2** Employee records table with avatar, role, today's attendance, lateness marks, and compliance/cert flags.
- **FR-HR-3 Onboard staff** — name, role, phone, next of kin, shift start, salary structure (base/transport/housing); created **Active** with compliance docs unchecked and a 1-year food-handler cert.
- **FR-HR-4 Employee file** — contact, compliance documents (guarantor / contract / ID card) with **file upload** (PDF / JPG / PNG,  $\leq 5$  MB), Replace and Download actions, upload date, filename and size; plus certification status, salary structure, last-5 attendance, disciplinary & commendation history.
- **FR-HR-5 Clock in / out** — clock-in computes lateness vs scheduled start.
- **FR-HR-6 Log incident** — type (Lateness, Customer complaint, Shortage, Misconduct, Damage, Commendation) + description + action; audit-logged.
- **FR-HR-7** A three-strike lateness rule surfaces a "warning due" flag at  $\geq 3$  late marks.
- **FR-HR-8 — Offboard staff:** an employee can be **Offboarded** with a reason — they no longer count in active staff, the roster, or payroll, but the record (attendance, disciplinary, salary history) is **retained on file, never deleted**. Offboarding is reversible via **Reactivate**.
- **FR-HR-9 — Status filter:** the records table hides offboarded staff by default; a toggle shows them (greyed) for audit and HR review.
- **FR-HR-10 — Weekly schedule:** each employee has a 7-day schedule (Mon–Sun) where each day is assigned a shift period — **Off**, **Full day**, **1st shift**, **2nd shift**, or **3rd shift**. The editor uses **per-day chip pickers with quick-fill presets** (Mon–Fri 1st shift, Mon–Sat Full day, etc.), colour-coded by shift, and a working-days summary — modeled on Deputy / When I Work. Rolled up into the HR Dashboard roster.



## 11.15 Payroll ( /payroll )

**Users:** owner, manager, hr.

- **FR-PAY-1** KPI tiles: On payroll, Monthly gross, **Pending bonus / deductions** (sums + count), Last run net.
- **FR-PAY-2 Run payroll** for a named period — generates a `Payslip` for every active employee at the branch with three sources of lines:
  1. **Statutory** (always present): PAYE 8% of gross, Pension 8% of base, NHF 2.5% of base.
  2. **Auto-generated itemised** lines (each with a written `reason`): lateness ( $\text{£}500 \times \text{late marks}$ , listing how many days), cash shortages from shift reconciliation (listing the shift IDs), welfare-advance instalments (labelled "*Instalment N of M · {welfareId}*").
  3. **Manual itemised** lines from the pending `PayrollAdjustment` queue — bonuses + deductions that a manager added between runs. Each consumed adjustment is marked with the run's id so it can't be double-counted.
- **FR-PAY-3 Pre-payroll adjustments** — a `PayrollAdjustment` queue on `/payroll` lets owner / manager / HR add **bonuses** (Performance bonus, Sales target, Overtime, Holiday bonus, Commendation, Allowance, Refund, Other) or **deductions** (Lateness, Cash shortage, Damage, Customer compensation, Uniform, Loan repayment, Other) — each requiring an **employee, category, amount, and a written reason**. The reason shows on the payslip; the audit log records the line as *Payroll bonus queued / Payroll deduction queued*. Removable until consumed by a run.
- **FR-PAY-4 Itemised payslip view** — earnings (basic + allowances → gross), then **Additions & bonuses** (each line shows category, italic reason, +~~N~~amount), then **Statutory deductions** (PAYE / Pension / NHF), then **Other deductions** (each line shows category, italic reason, −~~N~~amount, including any welfare-advance instalments), then Net pay. Printable via browser.
- **FR-PAY-5 Export bank schedule** as CSV: employee, role, gross, bonuses, deductions, net.
- **FR-PAY-6 Run history** of prior payroll runs.

## 11.16 Staff Welfare ( /welfare )

**Users:** owner, manager, hr, accountant.

Captures the informal-but-frequent staff support common in restaurant ops — medical bills, bereavement, family emergencies, education, weddings, salary advances, gifts. Distinct from petty cash (which covers business expenses, not staff care) and from payroll (which is regular salary).

- **FR-WEL-1** KPI tiles: Pending requests, Live (disbursed) count, **Advances outstanding** (~~£~~ to be recovered via payroll), Year-to-date paid.
- **FR-WEL-2 New request** — capture employee, category (Medical / Bereavement / Family emergency / Education / Wedding / Loan / salary advance / Gift / commendation / Other), amount, written reason. If the operator marks the request **repayable**, they pick a repayment plan of 1, 2, 3, 4, 6, 9, or 12 months; the modal previews the per-month instalment live.
- **FR-WEL-3** Lifecycle: **Pending** → **Approved** → **Disbursed** → **Closed / Repaying** → **Closed**, with **Rejected** as a terminal alternative carrying its own reason. HR / Manager / Owner approve; Accountant / Owner / Manager disburse.
- **FR-WEL-4 Auto repayment via payroll** — every `Repaying` request adds a labelled deduction line to its employee's next payslip ("*Welfare advance · Instalment N of M · {id} · {category}*"); the welfare record's `amountRepaid` increments, and on the final instalment the request auto-transitions to `Closed`.



- **FR-WEL-5 Audit** — every welfare action (request, approve, reject, disburse, close) logs to the audit trail under category **HR**, with the actor, amount, and reason in the detail.
- **FR-WEL-6 Pages segmented by status** — Pending, Approved, Live (Disbursed / Repaying), and History — so the operator sees the next action they need to take at the top.

### 11.17 Dispatch & Fleet ( /dispatch )

**Users:** owner, manager, cashier.

- **FR-DSP-1** KPI tiles: To dispatch, Out for delivery, COD outstanding, Delivered.
- **FR-DSP-2 Active deliveries** board — each delivery job (created from a POS delivery order) shows customer, address, phone, COD/prepaid, fee, age, assigned rider.
- **FR-DSP-3** Delivery lifecycle: **Preparing** → **Ready for pickup** → **Out for delivery** → **Delivered**.
- **FR-DSP-4 Assign rider** — hand a ready job to an internal bike or 3PL partner; COD jobs warn the assigner of the cash to collect.
- **FR-DSP-5 Fleet roster** — riders (internal bike / 3PL) with status, deliveries completed, fee revenue, and net P&L (bikes) or COD held (3PL).
- **FR-DSP-6 Add rider, clock in/out, log fleet expense** (fuel/repair/maintenance — reduces a bike's net P&L), and **settle COD** (a rider cannot clock out while holding unsettled COD cash).

### 11.17 Customers / CRM ( /customers )

**Users:** owner, manager.

- **FR-CRM-1** KPI tiles: Customers, VIPs, Open complaints, Branch mood (avg feedback rating).
- **FR-CRM-2 Golden-record database** keyed by **phone**; customers are auto-captured from takeout/delivery orders.
- **FR-CRM-3 Customer 360°** — tier control (New/Regular/VIP/Blacklisted), visits, lifetime value, average spend, last seen, favourite item, complaint and feedback history.
- **FR-CRM-4 Log feedback** — food/service/ambience star ratings + comment; a keyword sentiment classifier tags it; a low average ( $\leq 2$ ) or negative sentiment **auto-raises a complaint ticket**.
- **FR-CRM-5 Complaint tickets** — Open → In progress → Resolved; tickets open > 24h flagged as escalated.
- **FR-CRM-6 Marketing** — win-back list (not seen 30+ days) and birthdays-this-month list, each with a one-tap "contact" action that records the outreach.

### 11.18 Events / BEO ( /events )

**Users:** owner.

- **FR-EVT-1** KPI tiles: Live now, Bookings, Pipeline value, Avg guests.
- **FR-EVT-2 New booking** — name, schedule, venue, guests, package, contract value, deposit, and cost lines.
- **FR-EVT-3 Banquet Event Order (BEO)** view — guest/revenue/per-guest stats, cost breakdown, **event P&L** with margin %, deposit vs balance.
- **FR-EVT-4 Record deposit** against an outstanding balance; the first deposit confirms a pending booking.
- **FR-EVT-5** Status flow Deposit pending → Confirmed → Live → Completed; **print BEO**.

## 11.19 Analytics ( /reports )

**Users:** owner, manager.

- **FR-RPT-1** KPI tiles: Revenue, Gross margin (with COGS), Avg ticket, Items sold.
- **FR-RPT-2 Menu engineering matrix** — classifies items into Star / Puzzle / Plowhorse / Dog by margin vs popularity, with a recommendation per quadrant.
- **FR-RPT-3 Revenue by channel** and **Peak hours** (24-hour order histogram for staffing).
- **FR-RPT-4 Stock leakage** — variance loss + waste cost as a % of revenue.
- **FR-RPT-5 702 Morning Brief** — auto night-audit summary (sales, orders, top item, cash shortage, lateness, open complaints).
- **FR-RPT-6 Branch comparison** (owner only) — today's sales across all branches.
- **FR-RPT-7** Export menu-engineering analysis to CSV.

## 11.20 Audit Trail ( /audit )

**Users:** owner (group-wide), manager (own branch).

- **FR-ATR-1** KPI tiles: Events logged, Today, Sensitive actions, Staff acting.
- **FR-ATR-2** Filter by category (Sales, Inventory, Transfers, Procurement, Finance, Payroll, HR), free-text search, branch (owner), and a "sensitive only" toggle.
- **FR-ATR-3** Date-grouped timeline; each entry shows action, category, detail, actor, branch, time, amount, severity.
- **FR-ATR-4** Export the filtered log to CSV. Entries are immutable.

## 11.21 Alerts & Security ( /alerts )

**Users:** owner.

- **FR-ALT-1** KPI tiles: Critical, Warnings, Acknowledged, Active total.
- **FR-ALT-2 Live alert feed** derived from current-branch state: over-pours, low/out stock, expiring/expired batches; alerts can be viewed, acknowledged, and marked all read. Alerts are **grouped by section** — Main Store, Kitchen, Bar, Juice Bar — so each consuming area sees the items relevant to it (low stock by location; over-pours attributed to Bar; expiring batches grouped by the product's line). Sections with no active alerts are hidden.
- **FR-ALT-3** Alert detail explains the anomaly and recommends an action, deep-linking to Inventory.
- **FR-ALT-4 Security posture** checklist (informational): 2FA on admins, RBAC reviewed, audit-log streaming, <24h backup, idle auto-lock.

## 11.22 HR Dashboard ( /hr )

**Users:** owner, manager. **Purpose:** headcount, scheduling and compliance at a glance.

- **FR-HRDB-1 — KPIs:** Active staff, On shift today, Onboarded (30d), Offboarded (30d); plus Compliance gaps, Certs expiring and Late today panels.
- **FR-HRDB-2 — Today's roster:** active staff scheduled for today, grouped by shift period (Full day / 1st / 2nd / 3rd shift), with on-site indicator and late-mark flag.
- **FR-HRDB-3 — Weekly schedule grid:** rows = active staff, columns = Mon–Sun; each cell shows the assigned shift; today's column highlighted.

- **FR-HRDB-4 — Onboarding / offboarding history:** rolling 30-day lists of new hires and offboarded staff (records retained on file, never deleted).

## 12. Shared components & utilities

| Component / lib                               | Capability                                                                                                 |
|-----------------------------------------------|------------------------------------------------------------------------------------------------------------|
| <code>Modal</code> + <code>ModalButton</code> | Standard dialog: backdrop, Escape-to-close, scroll lock, sizes sm–xl, primary/ghost/danger buttons.        |
| <code>ImportModal</code>                      | Generic CSV import: template download, file upload or paste, live preview, tolerant parsing, result toast. |
| <code>ManagerApprovalModal</code>             | PIN-gated authorisation with optional reason and <code>ownerOnly</code> restriction.                       |
| <code>WasteModal</code>                       | Log inventory waste with cost calculation and staff/shift attribution.                                     |
| <code>ShiftBanner</code>                      | Clock-in (opening float) and clock-out (cash reconciliation + bartender bar count).                        |
| <code>lib/csv.ts</code>                       | RFC-4180-ish CSV parser with quoted-field handling and case-insensitive column lookup.                     |
| <code>lib/export.ts</code>                    | Client-side CSV generation and download.                                                                   |
| <code>lib/utils.ts</code>                     | <code>cn()</code> Tailwind class merge helper.                                                             |

## 13. Non-functional requirements

| #     | Category                              | Requirement                                                                                                                                 |
|-------|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| NFR-1 | Performance                           | First-load JS $\leq$ ~150 kB per route (current build is within this); interactions feel instant; KPIs derive client-side without blocking. |
| NFR-2 | Responsiveness                        | Fully usable on tablet and mobile; sidebar collapses to a drawer below <code>lg</code> . POS and KDS are touch-first.                       |
| NFR-3 | Availability<br><i>(production)</i>   | Front-of-house functions (POS, KDS) must tolerate intermittent connectivity — an offline/queue mode is required.                            |
| NFR-4 | Security <i>(production)</i>          | Hashed credentials, server-enforced RBAC, encrypted transport, session expiry, idle auto-lock, no secrets in client code.                   |
| NFR-5 | Auditability                          | Every money/stock-affecting action is logged immutably with actor, branch, time, and amount.                                                |
| NFR-6 | Data integrity<br><i>(production)</i> | Stock deduction, shift reconciliation, transfers, and payroll must be transactional and safe under concurrent access.                       |
| NFR-7 | Accessibility                         | Keyboard-operable controls, sufficient colour contrast, ARIA labels on icon-only buttons.                                                   |
| NFR-8 | Localisation                          | Currency is Nigerian Naira (₦); tax and statutory rates reflect Nigerian regulation; English UI.                                            |

| #      | Category        | Requirement                                                                                                          |
|--------|-----------------|----------------------------------------------------------------------------------------------------------------------|
| NFR-9  | Maintainability | Pages depend only on the <code>useStore()</code> hook surface; the data layer can be swapped without touching pages. |
| NFR-10 | Browser support | Modern evergreen browsers; <code>localStorage</code> required in the prototype.                                      |

## 14. Assumptions & constraints

- **A1** A single restaurant group; one hub + three branches. The design generalises to N branches but multi-tenant (multiple groups) is out of scope.
- **A2** Staff are trusted to the extent their role allows; the audit trail and approvals are the control, not technical lockdown (in the prototype).
- **A3** Nigerian regulatory context: 7.5% VAT, PAYE/Pension/NHF rates as coded. These must be configurable in production.
- **A4** All monetary values are whole Naira; no sub-unit (kobo) handling.
- **A5** The prototype assumes one device per role per branch; concurrent multi-device editing is a production concern.
- **C1** No backend, database, or real auth exists yet — see §15.
- **C2** Hardware (printers, drawers, terminals, scales) is not integrated; "print" uses the browser print dialog.

## 15. Current state — prototype limitations

The application is **functionally complete as a prototype** but carries deliberate shortcuts that must be resolved before production:

| Area                   | Current state                                                                              | Risk                                                     |
|------------------------|--------------------------------------------------------------------------------------------|----------------------------------------------------------|
| Persistence            | Browser <code>localStorage</code> only; per-device, clearable, single-user                 | No shared state; data loss; no real multi-device         |
| Authentication         | 4-digit PINs hardcoded in <code>STAFF_ROSTER</code> ; session in <code>localStorage</code> | Not secure; PINs visible in client source                |
| RBAC enforcement       | Sidebar links filtered only; routes not hard-gated                                         | Direct-URL access bypasses the access matrix             |
| Sentiment analysis     | Keyword classifier ( <code>sentimentOf</code> )                                            | Crude; misclassifies nuanced feedback                    |
| Alerts acknowledgement | Session-local; not persisted                                                               | Acknowledgements lost on refresh                         |
| Notifications          | Derived on render; "read" state session-local                                              | No real push; no history                                 |
| Reports/figures        | Some seeded shifts carry <code>seedSales</code> baselines                                  | Mixed seeded/real data until enough real activity exists |
| Printing & exports     | Browser print + client-side CSV                                                            | No formal document templates                             |

| Area        | Current state | Risk                                  |
|-------------|---------------|---------------------------------------|
| Concurrency | None          | Last-write-wins; no conflict handling |

## 16. Path to production (recommended roadmap)

### Phase 1 — Backend foundation

- Stand up a database and API; port every `useStore()` mutation to a server endpoint, preserving the hook's function surface so pages are untouched.
- Real authentication (hashed credentials, sessions/JWT), server-enforced RBAC per the §8.1 matrix, idle auto-lock.
- Move audit logging server-side and make it genuinely append-only.

### Phase 2 — Operational hardening

- Offline/queue mode for POS and KDS so trading survives connectivity loss.
- Real-time sync across devices (websockets) for tickets, tables, transfers, and alerts.
- Transactional stock deduction and shift reconciliation safe under concurrency.
- Hardware integration: receipt/kitchen printers, cash drawer, card terminal, weighing scale.

### Phase 3 — Intelligence & integrations

- Replace the keyword sentiment classifier with a proper model.
- Payment-gateway and bank integrations (card, transfer confirmation, payroll disbursement).
- Scheduled/emailed reports (the "702 Morning Brief" already has the UI hook).
- Configurable tax/statutory rates, categories, budgets, and approval tiers (admin settings).

### Phase 4 — Scale

- Multi-tenant support (multiple restaurant groups).
- Native mobile apps for managers/owners and for riders.
- Customer-facing ordering and loyalty.

## 17. Risks & open questions

| #  | Risk / question                                      | Notes                                                            |
|----|------------------------------------------------------|------------------------------------------------------------------|
| R1 | Routes are not role-gated server-side                | Must be closed before any real deployment; currently nav-only.   |
| R2 | Inventory <code>reorder</code> is a single par level | Production may need min/max and per-branch pars.                 |
| R3 | Transfers always originate from the hub              | Branch-to-branch transfers are not modelled — confirm if needed. |

| #  | Risk / question                                                                | Notes                                                          |
|----|--------------------------------------------------------------------------------|----------------------------------------------------------------|
| R4 | Payroll rates are hardcoded                                                    | Need an admin settings surface; rates change by regulation.    |
| R5 | Customer identity is phone-only                                                | Collisions/format variance possible; need normalisation rules. |
| R6 | No refund/partial-refund flow                                                  | VOIDS return stock but there is no post-payment refund path.   |
| Q1 | Should managers see other branches read-only?                                  | Today they are fully branch-locked.                            |
| Q2 | Are branch-to-branch (non-hub) transfers required?                             | Affects the transfer model.                                    |
| Q3 | What is the source of truth for cash movements between till, wallet, and bank? | Needs a cash-management spec.                                  |

## 18. Success metrics

| Metric                        | Definition                                                  | Why it matters                             |
|-------------------------------|-------------------------------------------------------------|--------------------------------------------|
| Stock leakage %               | $(\text{variance loss} + \text{waste}) \div \text{revenue}$ | Core value proposition — should trend down |
| Shift variance rate           | Share of shifts closing non-zero                            | Measures cash discipline                   |
| Over-pour incidents           | Negative Bar count variances per period                     | Bar control                                |
| Approval cycle time           | Request → decision for transfers/expenses                   | Manager responsiveness                     |
| Transfer dispute rate         | $\text{Disputed} \div \text{total transfers}$               | Supply-chain integrity                     |
| Menu mix health               | Share of revenue from Star items                            | Menu engineering effectiveness             |
| Gross margin %                | $(\text{Revenue} - \text{COGS}) \div \text{revenue}$        | Profitability                              |
| Daily active roles per branch | Roles signing in and transacting                            | Adoption                                   |

## 19. Appendix

### 19.1 Development login PINs (prototype only)

| Role      | Name      | PIN  | Lands on           |
|-----------|-----------|------|--------------------|
| owner     | Seun O.   | 0000 | /                  |
| manager   | Tunde A.  | 1111 | /manager-dashboard |
| cashier   | Ada O.    | 2222 | /cashier-home      |
| kitchen   | Amara K.  | 3333 | /kitchen-home      |
| bartender | Chukwu B. | 4444 | /bar-home          |

| Role        | Name     | PIN  | Lands on    |
|-------------|----------|------|-------------|
| storekeeper | Eze M.   | 5555 | /store-home |
| accountant  | Bola F.  | 6666 | /expenses   |
| hr          | Funke I. | 7777 | /hr         |

## 19.2 Glossary

| Term                            | Meaning                                                                   |
|---------------------------------|---------------------------------------------------------------------------|
| ROS                             | Restaurant Operating System                                               |
| Strong Room                     | The central warehouse hub; sole origin of stock transfers                 |
| Hub-and-spoke                   | Operating model: central hub supplies the branch "spokes"                 |
| Waybill                         | The document/record generated when a transfer is issued                   |
| Over-pour                       | A negative Bar stock variance — drink poured but not sold                 |
| Blind receiving                 | Receiving goods without seeing ordered quantities, to force honest counts |
| BEO                             | Banquet Event Order — the operating sheet for an event                    |
| Golden record                   | The single deduplicated customer record, keyed by phone                   |
| Imprest / float                 | A fixed petty-cash level a wallet is topped back up to                    |
| COD                             | Cash on delivery                                                          |
| Plowhorse / Star / Puzzle / Dog | Menu-engineering quadrants (popularity × margin)                          |
| Par / reorder level             | The stock level at or below which an item counts as Low                   |
| KDS                             | Kitchen Display System                                                    |

## 19.3 Module ↔ route ↔ source index

| Module                | Route              | Source file                    |
|-----------------------|--------------------|--------------------------------|
| Login                 | /login             | app/login/page.tsx             |
| Operations Overview   | /                  | app/page.tsx                   |
| Manager Dashboard     | /manager-dashboard | app/manager-dashboard/page.tsx |
| Cashier Home          | /cashier-home      | app/cashier-home/page.tsx      |
| Kitchen Home          | /kitchen-home      | app/kitchen-home/page.tsx      |
| Bar Home              | /bar-home          | app/bar-home/page.tsx          |
| Store Home            | /store-home        | app/store-home/page.tsx        |
| Front of House        | /pos               | app/pos/page.tsx               |
| Kitchen & Bar Display | /kitchen-bar       | app/kitchen-bar/page.tsx       |



| Module                  | Route            | Source file                  |
|-------------------------|------------------|------------------------------|
| Front of House on Shift | /cashier         | app/cashier/page.tsx         |
| Inventory               | /inventory       | app/inventory/page.tsx       |
| Stock Transfers         | /transfers       | app/transfers/page.tsx       |
| Procurement             | /purchase-orders | app/purchase-orders/page.tsx |
| Vendors                 | /vendors         | app/vendors/page.tsx         |
| Menu & Recipes          | /menu            | app/menu/page.tsx            |
| Expenses & Petty Cash   | /expenses        | app/expenses/page.tsx        |
| Staff (HR)              | /staff           | app/staff/page.tsx           |
| HR Dashboard            | /hr              | app/hr/page.tsx              |
| Payroll                 | /payroll         | app/payroll/page.tsx         |
| Dispatch & Fleet        | /dispatch        | app/dispatch/page.tsx        |
| Customers (CRM)         | /customers       | app/customers/page.tsx       |
| Events / BEO            | /events          | app/events/page.tsx          |
| Analytics               | /reports         | app/reports/page.tsx         |
| Audit Trail             | /audit           | app/audit/page.tsx           |
| Alerts & Security       | /alerts          | app/alerts/page.tsx          |
| Auth / session          | —                | lib/auth.tsx                 |
| Domain store            | —                | lib/store.tsx                |
| App shell               | —                | components/AppShell.tsx      |

## 20. Stakeholder review — decisions implemented in v1.1

Captured from the Tech4mation product review on **2026-05-22**. Every decision below has been **implemented in the application** and is reflected in the requirements above.

| #  | Decision                                                  | Rationale                                                                                                 | Where it landed                                                                        |
|----|-----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| D1 | Order <b>On hold</b> status — the <b>open-tab</b> model   | An order is placed and fires to the kitchen, but the tab stays open so the customer can keep adding to it | Order.status , recordSale(hold) , appendToOrder ; Open tab + amend flow (FR-POS-12/13) |
| D2 | An order is <b>complete only when payment is received</b> | "Sales" must mean money collected; a tab is closed deliberately at the end                                | closeOrder ; on-hold tabs excluded from revenue (FR-POS-14)                            |

| #   | Decision                                                                                         | Rationale                                                       | Where it landed                                                       |
|-----|--------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|-----------------------------------------------------------------------|
| D3  | Dashboard: daily sales, total orders, open bar/kitchen tickets, clickable low-stock notification | Confirmed the existing Overview design                          | Validated — Overview KPIs now count paid orders only                  |
| D4  | <b>Separate inventory</b> by section — Kitchen, Bar, Juice Bar                                   | Faster, independent audits; sections deplete at different rates | <code>Line</code> type + Inventory filter add "Juice Bar" (FR-INV-10) |
| D5  | Menu <b>groups</b> → <b>sub-items</b> ; <b>recent orders</b> visible                             | Confirmed the existing menu & feed design                       | Validated — no change needed                                          |
| D6  | Rename " <b>Point of Sale</b> " → " <b>Front of House</b> "                                      | "POS" reads as the card terminal; confused users                | Sidebar + page label; route <code>/pos</code> kept                    |
| D7  | Rename " <b>process order</b> " → " <b>Procurement</b> "                                         | Clearer for staff ordering inventory                            | Sidebar + page label; route <code>/purchase-orders</code> kept        |
| D8  | Rename " <b>Cashier &amp; Shifts</b> " → " <b>Front of House on Shift</b> "                      | Serves waiters and cashiers alike                               | Sidebar + page label; route <code>/cashier</code> kept                |
| D9  | <b>Shift history</b> with dates + date drop-down                                                 | Reconciliation needs a historical view                          | FR-SHF-6 — date column + filter on the ledger                         |
| D10 | <b>Configurable shift periods</b> — single or multiple shifts                                    | Branches operate differently                                    | FR-SHF-7 — period picker on clock-in (Full day / 1st / 2nd / 3rd)     |

## 20.1 Note on routes

The renames (D6–D8) change on-screen labels only. URL paths ( `/pos` , `/cashier` , `/purchase-orders` ) are unchanged, so existing links and bookmarks keep working.

## 21. Supply-chain & finance review — decisions implemented in v1.3

A further review refined the supply-chain and finance model. All decisions below are **implemented in the application**.

| #   | Decision                                                                                                           | Implementation                                                                                                                                                  |
|-----|--------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| D11 | The Strong Room <b>sells</b> stock to branches — a <b>receipt</b> is required                                      | Issued transfers produce a printable Strong Room issue receipt (FR-TRF-7)                                                                                       |
| D12 | Kitchen, Bar and Juice Bar each hold <b>their own inventory</b> , fed from the branch Main Store <b>on request</b> | Multi-location inventory + internal stock requests (FR-INV-11/12)                                                                                               |
| D13 | <b>No duplicate goods</b> across the Main Store and the sub-stores                                                 | One shared product catalogue; stock rows keyed by branch + location + SKU                                                                                       |
| D14 | Branch <b>purchase orders need management approval</b>                                                             | PO Pending Approval → Ordered / Rejected (FR-PO-8); destination is locked to the current branch — branches cannot request products for other branches (FR-PO-9) |
| D15 | Petty cash is <b>approved by an Accountant</b>                                                                     | New 7th <b>Accountant</b> role with its own login and finance workspace (FR-EXP-4)                                                                              |

## 21.1 The location model

Each branch has four stock locations — **Main Store, Kitchen, Bar, Juice Bar** (the Strong Room has only a store). Goods flow: **Strong Room** → **branch Main Store** (transfer + receipt) → **Main Store** → **Kitchen / Bar / Juice Bar** (internal stock request) → consumed by Front-of-House sales, which deduct each recipe ingredient from the sub-store matching its line. A product is defined once in the shared catalogue; each location simply holds its own count, so the same goods are never duplicated.

## 22. HR review — decisions implemented in v1.4

A further review formalised the staff lifecycle and scheduling.

| #   | Decision                                                                              | Implementation                                                                                                                                                                                                 |
|-----|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| D16 | Staff can be <b>onboarded</b> and <b>offboarded</b> — never deleted, records retained | <code>Employee.status = "Offboarded"</code> + <code>offboardEmployee / reactivateEmployee</code> mutations; attendance, disciplinary and salary history are kept on file. Offboard captures a reason and date. |
| D17 | A proper way to <b>schedule staffing on shift</b>                                     | <code>Employee.weeklySchedule</code> (Mon–Sun) per staff member, each day assigned a shift period (Off / Full day / 1st / 2nd / 3rd shift); edited from the employee file.                                     |
| D18 | An <b>HR dashboard</b>                                                                | New <code>/hr</code> page — KPIs, today's roster grouped by shift, a Mon–Sun schedule grid, and recent onboarding / offboarding (FR-HRDB-1.4).                                                                 |

## 24. Kitchen prep review — decisions implemented in v1.6

Refines how the Kitchen records goods and waste during prep.

| #   | Decision                                                                                                         | Implementation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----|------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| D23 | When stock arrives, the Kitchen logs it in <b>both units</b> (e.g. 10 pcs <b>and</b> ~30 kg of yam)              | <code>InventoryItem.altUnit</code> + <code>altOnHand</code> . Seeded for <b>Yam</b> (kg + pcs), <b>Plantain</b> (pcs + kg), <b>Tilapia</b> (pcs + kg). The <b>Adjust</b> → <b>Receive stock</b> dialog shows a second optional field "Also count in {altUnit}"; the inventory table displays "30 kg ≈ 10 pcs" on these items. Recipe deduction stays on the primary unit; the alt is a reference and is manually re-counted via Adjust.                                                         |
| D24 | After prep they <b>scale the waste</b> (e.g. plantain peels from 5 kg of plantain) and can <b>upload a photo</b> | <code>WasteEntry.photoName</code> / <code>photoDataUrl</code> . The Waste modal has a " <b>Take or upload photo</b> " affordance — on mobile it opens the device camera via <code>&lt;input type="file" accept="image/*" capture="environment"&gt;</code> ; on desktop it falls back to a file picker. A new " <b>Prep trim / peel</b> " reason is the default. Small photos store a data URL so the inventory waste log shows a 48 px thumbnail; larger photos store the filename as a record. |
| D25 | The Kitchen needs its own <b>Record waste</b> action                                                             | Kitchen home now has both <b>Request stock</b> and <b>Record waste</b> buttons (matches Bar home), opening the Waste modal with <code>location="kitchen"</code> .                                                                                                                                                                                                                                                                                                                               |

### 24.1 No-duplicates safeguard — added in v1.7

D13 ("no duplicate goods across the Main Store and the sub-stores") was already enforced by the **data model** — every stock row is keyed by `branch + location + SKU`, sub-stores reuse the catalogue's SKU when

stock is requested, and Strong Room transfers never create new product definitions. v1.7 closes the remaining **UI seam**:

- The **New product** button is only rendered on the **Main Store** tab. On Kitchen / Bar / Juice Bar tabs it is replaced by an inline hint: *"Stocked from the Main Store · products are defined there."*
- The **New product** dialog (formerly *New SKU*) does a **case-insensitive name lookup** against `store.inventory` as the user types. If the name already exists anywhere in the catalogue, the dialog:
  1. Shows an amber warning panel naming the existing SKU and its location (e.g. *"'Yam' is already in the catalogue · Tracked as KIT-002 in Kitchen Store · use Adjust → Receive stock to add more, or request it into a sub-store."*).
  2. Disables the **Add product** action.
- The same dialog now offers an **alt unit** dropdown at creation time, so dual-unit goods (yam in kg + pcs, plantain in pcs + kg, etc.) can be set up correctly from the start rather than retrofitted.

Net effect: there is no longer any path for an operator to accidentally create a second "Yam" record by typing it in again, on any branch or sub-store.

## 25. Customer accounts — wallet, house tab & statements (added in v1.13)

A real restaurant-CRM extension covering two distinct customer-money flows that are common in Lagos hospitality: **regulars who pre-fund a tab** (prepay → order against it) and **corporate / member accounts** that eat through the month and settle on a statement.

### 25.1 Two-ledger model

|                            | Wallet (prepaid)                     | House account (credit)                        |
|----------------------------|--------------------------------------|-----------------------------------------------|
| <b>Money flow</b>          | Customer → restaurant                | Restaurant extends → customer pays later      |
| <b>Balance meaning</b>     | What we owe the customer (liability) | What the customer owes us (A/R asset)         |
| <b>Risk</b>                | None                                 | Bad-debt — capped by <code>creditLimit</code> |
| <b>Settlement</b>          | Spend draws balance down to ₦0       | Statement issued → customer pays → balance ₦0 |
| <b>Per-customer fields</b> | <code>wallet: number</code>          | <code>credit, creditLimit: number</code>      |

Both ledgers coexist on the same `Customer` — a customer can prepay *and* charge to credit. `creditLimit === 0` disables house charges.

### 25.2 Data model

- `Customer.wallet` — prepaid balance,  $\geq 0$ .
- `Customer.credit` — outstanding owed,  $\geq 0$ .
- `Customer.creditLimit` — cap on `credit`; `0` = house charges disabled.
- `CustomerLedgerEntry` — append-only with `kind`  $\in$  { `wallet-topup`, `wallet-spend`, `wallet-refund`, `credit-charge`, `credit-payment`, `credit-writeoff` }, optional `orderId` / `invoiceId` links, and the staff who processed it.

- `CustomerInvoice` (statement) — period, issue date, due date, status ( `Draft` / `Sent` / `Partially Paid` / `Paid` / `Overdue` / `Void` ), array of `lines` (each citing the source ledger entry), `subtotal` , `paid` , and a `reminders[]` log.
- `InvoiceReminder` — kind ∈ { `pre-due`, `on-due`, `overdue-7`, `overdue-14`, `overdue-30` }, channel `Email` | `SMS` , sent-by + timestamp.

Schema bump: `v18` → `v19` .

## 25.3 Mutations (audit-logged)

- `topUpCustomerWallet` — increases wallet, logs `wallet-topup` .
- `spendCustomerWallet` — decreases wallet, fails if balance insufficient.
- `chargeCustomerAccount` — increases credit, fails if `creditLimit === 0` or `credit + amount > creditLimit` (manager `override` flag bypasses limit). Warning audit when crossing 80 % of the limit.
- `recordCustomerPayment` — decreases credit, applies to a chosen statement or the **oldest open one** by due date. Marks statement `Paid` / `Partially Paid` automatically.
- `setCustomerCreditLimit` — owner/manager/accountant only.
- `generateCustomerInvoice` — sweeps all *unbilled* credit charges (no existing invoice line links to them) in a period into a `CustomerInvoice` . ID format `INV-YYYY-MM-NNNN` .
- `sendInvoiceReminder` — logs reminder + bumps status ( `Draft` → `Sent` on first send; `Sent` → `Overdue` on overdue-{7,14,30}).
- `voidInvoice` — soft-deletes a statement (status `Void` ).

## 25.4 Entry points (zero new nav items)

1. `/customers` **index** — new KPI tiles **On customer wallets** and **Outstanding A/R**, plus a new **Account** column on the table showing per-row balance chips.
2. **Accounts receivable section** on `/customers` — A/R aging buckets ( `Current` / `1-30` / `31-60` / `61-90` / `90+` ), per-customer rows with limit usage bar and a one-click *Open account* link.
3. **Customer 360 modal** — now tabbed (**Profile** / **Account** / **Activity**). The *Account* tab shows two big balance cards (wallet · house) with inline actions: **Top up · Charge spend · Post charge · Record payment · Generate statement · Edit limit**. Below: list of statements + recent ledger entries with debit/credit icons.
4. **Invoice modal** — printable statement (browser print → PDF), full reminder cadence with status pips, reminder history.

## 25.5 Reminder cadence (industry standard for A/R)

| Kind                    | Trigger           | Tone                                    |
|-------------------------|-------------------|-----------------------------------------|
| <code>pre-due</code>    | 3 days before due | FYI                                     |
| <code>on-due</code>     | On the due date   | Friendly                                |
| <code>overdue-7</code>  | 7 days late       | Firm                                    |
| <code>overdue-14</code> | 14 days late      | Escalation                              |
| <code>overdue-30</code> | 30 days late      | <b>Final notice</b> , account suspended |

All reminders are *manually fired* from the invoice modal in the prototype ( `Email` | `SMS` ); a sent-reminder leaves an entry in the invoice's `reminders[]` log and an audit-log entry.

## 25.6 Safeguards

- Wallet **never goes negative** — `spendCustomerWallet` rejects insufficient-funds attempts.
- House charge **blocked at the limit** — toast: "Over credit limit — collect payment first or get a manager override". The mutation accepts an `override: true` argument that owner/manager UI can pass to bypass.
- A credit limit **cannot be set below the current owed amount** (would orphan unpaid debt outside the cap).
- Every wallet/credit movement and every statement action is **audit-logged** under the **Finance** category.

## 25.7 Roles

- **Owner / Manager / Accountant** — can top up, set limits, generate statements, record payments, send reminders. These are the only roles that see the management actions on the Customer 360 *Account* tab.
- **Cashier / Bartender** — will be able to spend wallets and post house charges from the **POS payment screen** (Phase 2 — see below). Cannot adjust limits or issue statements.

## 25.8 Phase 2 — POS payment integration

The data model and mutations are ready for the POS. Next-iteration scope:

- Add **Wallet** and **House charge** chips to the POS PaymentScreen `payMethods` strip.
- Gate the chips behind "order has a customer linked" (extend dine-in flow with customer linkage).
- On payment-complete, fire `spendCustomerWallet` / `chargeCustomerAccount` against the new order's `orderId`, leaving a linked ledger entry.
- Surface wallet balance + remaining credit beside the customer's name in the cart header.

Until then, the **Post charge** action on the customer modal simulates exactly what the POS will do, so the wallet/credit/invoice flow is fully testable end-to-end today.

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## 26. Whole-product audit & onboarding roadmap (added in v1.17)

A thorough audit of the working prototype against the industry-standard restaurant-OS feature set, with each gap categorised, prioritised, and assigned to a phase. The implementation already closes the highest-priority items; the remainder is the roadmap.

### 26.1 Rider & fleet (mostly **CLOSED** in v1.17)

| Gap                                                         | Status   | Notes                                                                                                                                                 |
|-------------------------------------------------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Rider has identity (phone, next-of-kin, hire date, licence) | ✅ Closed | New required + optional fields on <code>Rider</code> . The add-rider modal asks for the full set on internal bikes and only contact for 3PL partners. |
| Bike plate, make, acquisition cost                          | ✅ Closed | Bike metadata on <code>Rider</code> ; plate is visible on every dispatch card; acquisition cost seeds the P&L.                                        |

| Gap                                                   | Status            | Notes                                                                                                                                                                                                                                                                                   |
|-------------------------------------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bike P&L (money in vs out)                            | ✅ Closed          | New <code>FleetTxn</code> append-only ledger with kinds <code>purchase / fuel / maintenance / fine / delivery-fee / income / expense</code> . <code>completeDelivery</code> auto-credits the fee. Rider 360 shows <code>Income · Outflow · Net P&amp;L</code> with a date-range filter. |
| Rider performance (deliveries, on-time, avg duration) | ✅ Closed          | Rider 360 computes deliveries-in-window, on-time rate ( $\leq 45$ min from assignment to delivered), and average duration.                                                                                                                                                              |
| COD held vs settled                                   | ✅ Already in v1.0 | <code>settleCOD</code> was already wired; the dispatch card calls it out in red until settled.                                                                                                                                                                                          |
| Rider attendance / scheduling                         | 📅 Roadmap         | Today rider clock-in/out is just status. Production: link <code>Rider</code> to an <code>Employee</code> record for full scheduling.                                                                                                                                                    |
| Bike depreciation                                     | 📅 Roadmap         | Acquisition is treated as a single outflow. Production: straight-line depreciation over the asset's useful life (typically 3 years for delivery bikes).                                                                                                                                 |
| Rider commission / payout                             | 📅 Roadmap         | Currently fees credit the bike — not the rider personally. Production needs a commission model (e.g. ¥300/delivery + base) and a payable to the rider.                                                                                                                                  |

## 26.2 Delivery flow (mostly **CLOSED** in v1.17)

| Gap                                                                  | Status    | Notes                                                                                                                                                                    |
|----------------------------------------------------------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| New cashier needs the address of a returning customer without asking | ✅ Closed  | Phone-first lookup in the POS Takeout/Delivery modal with live CRM suggestions and a "Returning customer" pill ( <code>tier · visits · lifetime · last address</code> ). |
| Customer's address persists across orders                            | ✅ Closed  | <code>Customer.address</code> added to the golden record. New / changed addresses save back on submit.                                                                   |
| Optional landmark                                                    | ✅ Closed  | <code>Customer.addressLandmark</code> field; not exposed in UI yet — surfaces in next iteration.                                                                         |
| Customer search by name (not just phone)                             | ✅ Closed  | Suggestions match on either.                                                                                                                                             |
| Address book — multiple addresses per customer                       | 📅 Roadmap | Today one canonical address per customer. Industry standard for chain delivery is <i>home / office / saved</i> — modelled as an array.                                   |
| Geocoding & map                                                      | 📅 Roadmap | Address is free text. Production: optional pinned location for the rider.                                                                                                |
| Delivery time estimates                                              | 📅 Roadmap | Today the rider hits "Delivered" manually; no ETA exposed.                                                                                                               |
| Order tracking link for the customer                                 | 📅 Roadmap | Production would send the customer a status link.                                                                                                                        |



## 26.3 Floor plan / tables (NEW in v1.17)

| Gap                                | Status    | Notes                                                                                                                                                                                                                                          |
|------------------------------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tables are seeded — never editable | ✅ Closed  | New <code>/settings</code> page with full add / rename / re-seat / delete (delete blocked while occupied or reserved).                                                                                                                         |
| Tables grouped by zone             | ✅ Closed  | Settings → Floor plan groups by zone, suggests existing zones as datalist.                                                                                                                                                                     |
| Tables are <i>per-branch</i>       | 📅 Roadmap | <code>TableRec</code> doesn't carry a <code>branch</code> today — the seeded set is shared across all branches. Production: add <code>branch</code> to <code>TableRec</code> and scope all reads. <b>HIGH PRIORITY</b> — listed under Phase 2. |
| Drag-and-drop floor plan editor    | 📅 Roadmap | Settings is list-based; a visual editor (à la SevenRooms / OpenTable) is a nice-to-have.                                                                                                                                                       |
| Combine / split tables             | 📅 Roadmap | Industry standard for groups of 10+ — needs a join-tables flow.                                                                                                                                                                                |

## 26.4 Roles & role definitions (Roadmap)

| Gap                                                                                     | Status    | Notes                                                                                                                                                                                                                                   |
|-----------------------------------------------------------------------------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Roles are hardcoded ( <code>STAFF_ROSTER</code> + <code>StaffRole</code> literal union) | 📅 Roadmap | Today the eight roles (owner / manager / cashier / kitchen / bartender / storekeeper / accountant / hr) are baked into TypeScript. Production must let the owner <b>create custom roles</b> and define the permissions matrix per role. |
| PIN-based auth is dev-only                                                              | 📅 Roadmap | Already called out in §15 / §16. Production: phone or email + OTP / SSO.                                                                                                                                                                |
| Role permissions in code, not data                                                      | 📅 Roadmap | Each page hard-codes role gates. Production: a permissions table per role, role-permission joins, and a UI to edit them.                                                                                                                |
| Cross-branch staff assignment                                                           | 📅 Roadmap | One staff member is currently bound to one branch. Multi-branch managers (covering Ikoyi + Agungi) need a many-to-many.                                                                                                                 |

## 26.5 Restaurant onboarding (multi-tenant — Roadmap)

The current prototype is single-tenant (Haven 11 / 702 ROS hardcoded). For a productised offering serving multiple restaurant groups, an onboarding flow is needed. Phased model:

### Phase A — Self-serve sign-up (Day 0)

1. Owner signs up with email + phone. Identity verification.
2. Restaurant entity created (name, type — restaurant / lounge / hybrid / nightclub).
3. Default chart of accounts seeded, default categories, default reminder cadence.

### Phase B — Branch setup (Day 1)

1. Add branches (name, address, kind: hub / spoke).
2. For each branch: tables (via `/settings` — already shipped), zones, opening hours.
3. Define shift periods (already configurable per branch — see v1.1).

### Phase C — Catalogue (Day 2-3)

1. Bulk-import inventory via the existing **CSV importer** (already shipped — accepts SKU / name / category / line / on-hand / unit / reorder / cost).
2. Bulk-import menu items + recipes.
3. Define product categories (already manageable — v1.10).

### Phase D — Team (Day 3-7)

1. Onboard staff via the existing `/staff` page (already supports compliance docs — v1.5).
2. Assign roles + branches.
3. Set weekly schedule.

### Phase E — Suppliers & money

1. Add vendors with payment terms.
2. Set petty-cash float per branch.
3. Set tax / VAT rate (currently hardcoded at 7.5% — needs to be a setting).

### Phase F — Go live

1. Test order from cashier → kitchen → completed.
2. First real order.

The UI to drive Phase A → F is the **next-major-phase deliverable**. Today an admin who knows the data model can complete all steps via the existing screens (which is why each step's *destination screen* is already shipped). What's missing is the **guided checklist** that walks an owner through them.

## 26.6 Other gaps surfaced by the audit

| Area                                         | Gap                                                                                                                       | Priority                                                        |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| Tax / VAT                                    | 7.5% rate hardcoded; should be a branch-level setting                                                                     | <b>High</b> — easy fix                                          |
| Receipts                                     | No printer integration                                                                                                    | High — needs ESC/POS driver + USB/Bluetooth                     |
| Cash drawer                                  | No drawer-open event on payment                                                                                           | Medium — needs hardware integration                             |
| Reservations                                 | <code>TableRec.reservation</code> is a string note; needs a <code>Reservation</code> entity with party size, time, status | Medium                                                          |
| Recurring expenses (rent, payroll)           | Today expenses are one-off                                                                                                | Medium                                                          |
| Multi-currency                               | ¥ hardcoded; export operations may need USD/EUR                                                                           | Low                                                             |
| Audit trail filter by date range             | Audit log has no date filter yet                                                                                          | Medium — <code>DateRangePicker</code> exists, just needs wiring |
| Reports filter by date range                 | Same                                                                                                                      | Medium                                                          |
| Transfer / PO / Expense filter by date range | Same                                                                                                                      | Medium                                                          |
| Receipt-printer test page                    | None                                                                                                                      | Low — Phase B of onboarding                                     |

| Area                                     | Gap               | Priority                                                       |
|------------------------------------------|-------------------|----------------------------------------------------------------|
| Tax-receipt compliance (FIRS in Nigeria) | Not modelled      | High for production                                            |
| Backup / restore                         | localStorage only | <b>Critical</b> for production — needs server-side persistence |
| Multi-device sync                        | None              | Critical for production                                        |
| Offline mode                             | None              | High for Nigerian power-reliability reality                    |

## 26.7 What was closed this turn — recap

- **Rider profile + bike P&L + performance** — full identity record, append-only fleet ledger, auto-credited delivery fees, Rider 360 modal.
- **Customer-aware delivery flow** — phone-first lookup, autofill, persisted address — a new cashier never has to ask a regular for their address.
- **Floor-plan management** — `/settings` page lets owner/manager add / rename / delete tables grouped by zone.
- **Audit document** — this section, surfacing every remaining gap with priority and phase.

The remaining gaps (per-branch tables, custom roles, multi-tenant onboarding flow, hardware integration, server-side persistence) are explicitly tracked here and will be picked up in priority order.

## 26.8 Role permissions matrix (effective in v1.20)

Codifying what each role can / cannot do — industry standard for Toast / Square / Lightspeed. Production must move this from code-hardcoded gates to a permissions table editable from `/settings`.

| Action                                  | Owner | Manager | Cashier  | Bartender | Kitchen | Storekeeper | Accountant | HR |
|-----------------------------------------|-------|---------|----------|-----------|---------|-------------|------------|----|
| Take orders at POS                      | ✓     | ✓       | ✓        | ✓         |         |             |            |    |
| Take payment (cash/card/transfer/split) | ✓     | ✓       | ✓        | ✓         |         |             |            |    |
| Apply discount > ₦500 (needs override)  | ✓     | ✓       | with PIN | with PIN  |         |             |            |    |
| Void order                              | ✓     | ✓       | with PIN | with PIN  |         |             |            |    |
| Open/close shift                        | ✓     | ✓       | ✓        | ✓         |         |             |            |    |
| Mark ticket Ready (Kitchen/Bar)         | ✓     | ✓       |          | ✓         | ✓       |             |            |    |
| Mark ticket Served (waiter)             | ✓     | ✓       | ✓        | ✓         |         |             |            |    |
| Hire internal rider                     | ✓     | ✓       |          |           |         | ✓           |            |    |
| Add 3PL partner                         | ✓     | ✓       |          |           |         | ✓           |            |    |
| Assign delivery to rider                | ✓     | ✓       | ✓        |           |         | ✓           |            |    |

| Action                                  | Owner | Manager   | Cashier   | Bartender | Kitchen | Storekeeper | Accountant | HR |
|-----------------------------------------|-------|-----------|-----------|-----------|---------|-------------|------------|----|
| Mark delivery complete                  | ✓     | ✓         | ✓         |           |         | ✓           |            |    |
| Log fleet expense                       | ✓     | ✓         |           |           |         | ✓           |            |    |
| Settle COD                              | ✓     | ✓         |           |           |         | ✓           |            |    |
| Request petty cash                      | ✓     | ✓         | ✓         | ✓         |         | ✓           | ✓          |    |
| Approve petty cash                      | ✓     |           |           |           |         |             | ✓          |    |
| Disburse petty cash                     | ✓     |           |           |           |         |             | ✓          |    |
| Top up petty-cash float                 | ✓     |           |           |           |         |             | ✓          |    |
| Reconcile (return change + receipt)     | ✓     | requester | requester | requester |         | requester   | ✓          |    |
| Receive stock (Strong Room)             | ✓     | ✓         |           |           |         | ✓           |            |    |
| Approve transfer (Strong Room → branch) | ✓     | ✓         |           |           |         |             |            |    |
| Issue transfer                          | ✓     | ✓         |           |           |         | ✓           |            |    |
| Receive transfer at branch              | ✓     | ✓         |           |           |         | ✓           |            |    |
| Internal stock request                  | ✓     | ✓         |           | ✓         | ✓       | ✓           |            |    |
| Record waste                            | ✓     | ✓         |           | ✓         | ✓       | ✓           |            |    |
| Manage categories                       | ✓     | ✓         |           |           |         |             |            |    |
| Manage menu / recipes                   | ✓     |           |           |           |         |             |            |    |
| Raise PO                                | ✓     | ✓         |           |           |         | ✓           |            |    |
| Approve PO                              | ✓     | ✓         |           |           |         |             |            |    |
| Mark PO paid                            | ✓     |           |           |           |         |             | ✓          |    |
| Customer wallet top-up / charge         | ✓     | ✓         |           |           |         |             | ✓          |    |
| Set customer credit limit               | ✓     | ✓         |           |           |         |             | ✓          |    |
| Generate / send statement               | ✓     | ✓         |           |           |         |             | ✓          |    |
| Record customer payment                 | ✓     | ✓         |           |           |         |             | ✓          |    |
| Hire / off-board staff                  | ✓     | ✓         |           |           |         |             |            | ✓  |
| Edit schedule                           | ✓     | ✓         |           |           |         |             |            | ✓  |
| Run payroll                             | ✓     | ✓         |           |           |         |             |            | ✓  |

| Action                                     | Owner | Manager | Cashier | Bartender | Kitchen | Storekeeper | Accountant | HR |
|--------------------------------------------|-------|---------|---------|-----------|---------|-------------|------------|----|
| Queue payroll adjustment (bonus/deduction) | ✓     | ✓       |         |           |         |             |            | ✓  |
| Approve welfare request                    | ✓     | ✓       |         |           |         |             |            | ✓  |
| Disburse welfare                           | ✓     | ✓       |         |           |         |             | ✓          |    |
| View Financials (P&L · Balance Sheet)      | ✓     | ✓       |         |           |         |             | ✓          |    |
| Manage floor plan / tables                 | ✓     | ✓       |         |           |         |             |            |    |
| View Audit Trail                           | ✓     | ✓       |         |           |         |             | ✓          | ✓  |
| Roam branches                              | ✓     |         |         |           |         |             | ✓          | ✓  |

**Industry alignment:** matches the role split at Toast, Square for Restaurants, Lightspeed. The single Nigerian-specific addition is **Welfare disbursement** being shared between Accountant and Manager (in larger Western groups, that's HR's call; in our context, the cash custodian needs to be in the loop).

## 23. UX & HR review — decisions implemented in v1.5

A UX-quality pass on the HR experience, applying familiar patterns from established platforms.

| #   | Decision                                                                                 | Implementation                                                                                                                                                                                                                                                                                                                                                            |
|-----|------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| D19 | <b>HR should have their own login</b>                                                    | New 8th <b>HR Officer</b> role (Funke I., PIN 7777 , roaming) with a dedicated workspace — HR Dashboard, Staff, Payroll, Audit Trail. Adds a teal/indigo identity at the PIN pad.                                                                                                                                                                                         |
| D20 | <b>Upload the documents staff bring</b> (guarantor's form, etc.) — not just tick a box   | <code>Employee.compliance</code> is now three <code>ComplianceDoc</code> objects (guarantor / contract / ID card). The employee file shows each as a <b>file row</b> : status icon, filename, size, upload date, and Upload / Replace / Download actions. Accepts PDF / JPG / PNG up to 5 MB; small files store a data URL so they can be downloaded again from the demo. |
| D21 | The schedule editor should be <b>better</b> — <b>inspired by what other platforms do</b> | Redesigned with <b>quick-fill presets</b> (Mon–Fri 1st shift, Mon–Sat Full day, Tue–Sat 2nd shift, etc.) plus per-day <b>chip pickers</b> colour-coded by shift period (Full day = green, 1st = sky, 2nd = amber, 3rd = violet), with a working-days summary — modeled on Deputy / When I Work / 7shifts.                                                                 |
| D22 | <b>Reduce clutter</b> — follow UX patterns users already know                            | /staff replaces its dense table with a <b>list-row pattern</b> (avatar + name + role + status chip + key issues + chevron), like contact/people lists in Linear, GitHub and iOS Settings. /hr collapses three separate alert cards into one <b>alerts strip</b> , simplifies the today's-roster cards, and tightens the weekly grid colour scheme.                        |

*End of document.*